FS-CGNet: Frequency Spectral–Channel Fusion and Cross-Scale Global Aggregation Network for Hyperspectral Image Classification

Xiaoqing Wan, *Member, IEEE*, Feng Chen, Dongtao Mo, Hui Liu, Zhize Li, and Kun Hu

***Abstract*—Hyperspectral image (HSI) classification is a challenging task due to the high-dimensional attribute of the data and the scarcity of labeled samples. Many existing models struggle to simultaneously capture both local structural features and global patterns, limiting their ability to handle complex HSI tasks to some extent. Furthermore, they often fail to efficiently process multiscale and multifrequency information, resulting in suboptimal performance. To address these challenges, we propose FS-CGNet, a novel framework that integrates frequency spectral–channel fusion and cross-scale global aggregation networks for HSI classification. FS-CGNet introduces three key components: 1) a frequency-domain spectral feature extraction module (FDSFM) that utilizes techniques like fast Fourier transform (FFT) to capture multiscale and multifrequency information, improving the ability to discriminate local and global features; 2) a multiscale spectral–channel fusion attention module (MSFAM) that adaptively emphasizes critical spectral channels, enhancing classification accuracy and robustness; and 3) a cross-scale global aggregation module (CGAM) that aggregates features from multiple scales, ensuring optimal utilization of cross-scale information and improving overall classification performance. Comprehensive experiments on four benchmark HSI datasets (Pavia University, Salinas, WHU-Hi-LongKou, and Xiong An) demonstrate that FS-CGNet achieves the state-of-the-art performance, comparable to several leading methods in the field. The code of this work is available at https://github.com/wanxiaoqing/FS-CGNet**

***Index Terms*—Attention mechanism, classification, fast Fourier transform (FFT), hyperspectral image (HSI).**

Received 13 April 2025; revised 17 August 2025; accepted 13 September 2025. Date of publication 17 September 2025; date of current version 29 September 2025. This work was supported in part by the Scientific Research Fund of Hunan Provincial Education Department under Grant 23B0666, in part by the Science and Technology Innovation Program of Hunan Province under Grant 2016TP1020, in part by the 14th Five-Year Plan Key Disciplines and Application-Oriented Special Disciplines of Hunan Province under Grant Xiangjiaotong [2022] 351, in part by the Science Foundation of Hengyang Normal University under Grant 2022QD07, and in part by Hengyang City Guidance Plan Project under Grant 202323036897. *(Corresponding author: Xiaoqing Wan.)*

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Digital Object Identifier 10.1109/TGRS.2025.3610848

I. Introduction

HYPERSPECTRAL image (HSI) classification has emerged as a vital research area in remote sensing, with growing importance in diverse applications such as natural resource management [1], disaster response [2], [3], precision agriculture [4], [5], [6], and land use monitoring [7], [8], [9]. By capturing images in numerous narrow, continuous spectral bands, HSI provides in-depth spectral–spatial information that helps in distinguishing subtle variations between materials and objects in the scene. However, HSI classification is often hindered by challenges such as high-dimensional data [10], excessive redundancy [11], and the difficulty in obtaining enough labeled samples for training [12]. These issues complicate the process of accurately classifying land cover types, thereby limiting the performance and effectiveness of classification models.

In order to tackle the challenges in HSI classification, numerous machine learning (ML) [13] techniques have been introduced. Initially, many methods were based on classical ML algorithms, such as support vector machines (SVMs) [14], [15], [16], *k*-nearest neighbors (K-NN) [17], [18], and random forests (RFs) [19], [20]. These methods focus on classifying pixels based on predefined features or simple models, with feature extraction often being a crucial step. While these models can yield reasonable performance in certain cases, they face significant limitations when dealing with the high-dimensional nature of HSI data. Specifically, traditional ML techniques struggle with issues like the "curse of dimensionality [21]," where the number of features exceeds the number of training samples, resulting in overfitting [22]. Moreover, these methods rely heavily on handcrafted features and domain knowledge, making them less adaptive to complex and large-scale datasets.

In response to the limitations of traditional ML approaches in HSI classification, deep learning (DL) techniques [23], [24] have emerged as a promising alternative. Unlike conventional ML methods, DL algorithms have the ability to automatically learn complex, hierarchical data representations through multiple layers, which enables them to capture intricate patterns in high-dimensional feature spaces. This feature extraction capability, which eliminates the need for manual engineering, enhances their flexibility and effectiveness in handling large-scale, complex datasets. Among various DL architectures, convolutional neural networks (CNNs) [25], [26], [27] have gained significant attention in HSI classification due to their

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strength in capturing spatial relationships and structures in image data. CNNs have not only shown remarkable performance in object detection [28] and semantic segmentation [29], [30], [31] tasks but also are widely used in HSI classification tasks, where CNN-based models are capable of effectively identifying objects and understanding the spatial context of images. CNNs are particularly effective in learning local spatial patterns and are often combined with spectral information to improve classification accuracy. However, one challenge with CNNs in HSI classification is their limited capacity to fully utilize the spectral dimension of hyperspectral data. To address the limitation of CNNs in fully exploiting the spectral dimension of hyperspectral data for HSI classification, frequency-domain analysis has emerged as a promising solution. Traditional spatial-domain-based methods typically focus on pixel-wise features and may not fully capture the spectral information or its complex interactions with spatial structures. In contrast, frequency-domain analysis decomposes the signal into frequency components, allowing for a more comprehensive exploration of spectral patterns and their interactions with spatial structures. Additionally, frequency-domain analysis offers clear advantages in terms of noise robustness and computational efficiency, providing a more robust and efficient solution for complex HSI classification tasks. This approach also presents new avenues for enhancing DL techniques. More sophisticated architectures, such as 3-D CNNs [32], [33] and recurrent neural networks (RNNs) [34], [35], have been introduced. For example, 3-D CNNs operate simultaneously across both the spatial and spectral dimensions, enabling the model to capture spatial features and spectral correlations in an integrated manner. On the other hand, RNNs, especially long short-term memory (LSTM) networks [36], [37], are particularly effective for processing sequential data and can model temporal dependencies in hyperspectral data, which proves beneficial in dynamic settings or for time-series analysis. Additionally, some approaches have incorporated attention mechanisms [38], [39], enabling the model to focus on important regions or spectral bands. This selective attention improves the performance by emphasizing the most relevant features. Building on these advancements, researchers have continued to explore novel methods to enhance HSI classification. Dong et al. [40] proposed a weighted feature fusion method using CNN and graph attention networks (WFCG), which effectively combines features from both models to improve HSI classification performance. To address challenges related to uneven pixel distribution and inherent spectral correlation in HSI data, Pu et al. [41] introduced a multilevel feature network and spectral–spatial attention model (MFNSAM) that effectively bridges the gap between multilevel features. Yang et al. [42] proposed the cross-attention spectral–spatial network (CASSN). This innovative approach introduces a cross-spectral component that combines both local and global pixel spectra to compute band weights, effectively reducing the redundancy of spectral bands. Additionally, the method incorporates a cross-spatial attention component, which, driven by the target pixels, generates spectral–spatial features from HSI patches, enhancing the model's ability to capture spatial and spectral correlations more effectively.

Building upon the progress made with CNN in HSI classification, Transformer models [43], [44], [45], [46] have recently emerged as a highly effective alternative for this task. Initially developed for natural language processing (NLP) [47], Transformer models have demonstrated remarkable potential in managing sequential data and capturing long-range dependencies through their self-attention mechanisms. A significant advantage of Transformer models is their ability to simultaneously model both spatial and spectral features, eliminating the need for traditional convolutional or recurrent layers. Although the Transformer framework can represent high-level semantic features well, the computing cost increases significantly with the increase in layers. Therefore, Sun et al. [48] introduced a spectral–spatial feature tokerization Transformer (SSFTT) approach, designed to effectively capture both spectral–spatial features and high-level semantic representations. In a similar vein, Tu et al. [49] proposed a converter architecture that relies on a local semantic feature aggregation Transformer (LSFAT), which enhances the converter's ability to capture long-range dependencies across multiscale features more efficiently. This architecture also mitigates the computational burden typically associated with learning multiscale features. Additionally, Xu et al. [50] presented a double-branch convolutional Transformer network (DBCTNet), which integrates multiscale spectral feature extraction with a convolutional Transformer module. This design addresses the dual challenges of high computational complexity and the insufficient extraction of multispectral dimensional information in HSI classification tasks. Sun et al. [51] proposed a memory-augmented spectral–spatial Transformer (MASSFormer). By using memory marker and memory enhanced converter encoder module, MASSformer effectively overcomes the problems of information loss and insufficient mixing of spectral–spatial information in HSI by existing converter-based classification methods, thus improving the classification performance. Roy et al. [52] introduced the morphological Transformer (morphFormer), which combines learnable spectral and spatial morphological convolution operations with the attention mechanism. This integration aims to strengthen the interaction between spectral and spatial features, thereby improving the performance of HSI classification. On the other hand, Mei et al. [53] proposed a group-aware hierarchical Transformer (GAHT) specifically designed for HSI classification. The GAHT model leverages a grouped pixel embedding module that directs multihead self-attention toward localized spatial–spectral contexts. Additionally, it employs a hierarchical structure to enhance classification accuracy while minimizing the number of parameters required. In order to overcome the computational complexity and feature fusion problems of Transformer model when processing HSIs, Mamba [54], [55] model proposes a more efficient and multiscale adaptive solution. He et al. [56] proposed a 3DSS-Mamba framework for HSI classification that enhances global spectral–spatial relationship modeling with improved computational efficiency through a 3-D-spectral–spatial token generation and selective scanning mechanism. Sheng et al. [57] proposed the DualMamba framework for HSI classification that efficiently combines global spectral–spatial features from

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a lightweight parallel Mamba and CNN architecture, achieving superior accuracy with reduced computational complexity and model parameters. Yang et al. [58] proposed GraphMamba, an innovative HSI classification framework that combines efficient spectral and spatial information extraction through the HyperMamba module and SpatialGCN, enhancing computational efficiency and context awareness while improving classification performance.

However, despite CNN's remarkable success in many visual tasks, particularly in HSI classification and object recognition, it still has some limitations. CNN is effective at capturing local features and extracting them through convolutional layers, but it generally requires higher computing demands as the layer's number increases. In addition, the localized characteristic of CNN makes it less effective at capturing global information, which is crucial for HSI classification, where the spectral and spatial context is often interrelated and needs to be considered holistically. In contrast, Transformer, through its self-attention mechanism, can effectively capture global context and excels at handling long-range dependencies, especially in tasks like HSI classification. However, Transformer has extremely high computational demands, particularly when processing large-scale observations in remote sensing scenarios, where its computational complexity can lead to significant training and inference costs. Mamba, as an emerging DL model, aims to improve computational efficiency and optimize resource usage, reducing computational burdens to some extent, making it suitable for tasks that require considerable computational resources, such as HSI classification. However, Mamba also has certain limitations. Despite its linear complexity, the Mamba method can be resource-intensive and time-consuming for high-dimensional data due to multiple feature extraction modules. Additionally, its higher memory footprint compared to CNN and Transformer may become a bottleneck in resource-constrained environments. Overall, although these models perform prominently in the HSI classification task, the demand for computational resources remains a significant challenge in large-scale hyperspectral datasets. Therefore, to expand their feasibility in practical applications, efficient dimensionality reduction and optimization techniques can be adopted to reduce computational burdens in HSI classification.

In the current study, while many models successfully extract spectral–spatial features from HSI, they often struggle to fully capture local structural features and global patterns due to the high-dimensional nature of the data. Moreover, the lack of sufficient labeled samples in hyperspectral datasets further complicates the task, making it challenging for existing models to achieve robust and accurate classification. Furthermore, many current models fail to efficiently process multiscale and multifrequency information, leading to suboptimal performance when handling complex HSI tasks. To address these challenges, this article introduces a novel approach with the following main contributions.

1) A novel frequency-domain spectral feature extraction module (FDSFM) is proposed. By utilizing the fast Fourier transform (FFT), multiscale and multifrequency information is extracted from HSIs. This approach significantly enhances the ability to process high-dimensional data, particularly improving the discrimination capability in capturing local structural features and global patterns within HSIs.
2) A multiscale spectral–channel fusion attention module (MSFAM) is designed. This module fuses spectral information across different scales and dynamically enhances the importance of key features through an attention mechanism. This allows the model to adaptively focus on the most critical spectral channels for classification tasks. This fusion and weighting mechanism effectively improves classification accuracy and robustness.
3) A cross-scale global aggregation module (CGAM) is built to further enhance feature representation by aggregating information from multiple scales. This module ensures that features from different scales are fully utilized in the final network output, optimizing the classification task from a global perspective. The aggregation of cross-scale information effectively reduces the misclassification of pixels, leading to improved classification accuracy, especially in complex HSI classification tasks.
4) We conducted experiments using four widely recognized benchmark HSI datasets, Pavia University, Salinas, WHU-Hi-LongKou, and Xiong An. The results show that the proposed network performs on par with several leading state-of-the-art methods in the field.

II. Related Work

In this section, we provide a detailed and comprehensive overview of the frequency spectral–channel fusion and cross-scale global aggregation network (FS-CGNet) approach, explaining its complex architecture in depth. This includes an in-depth explanation of the FDSFM, the MSFAM, and the CGAM, all of which are visually illustrated in Fig. 1.

Although existing HSI classification methods have made progress in capturing local details, global structures, and frequency-domain information, they often struggle to simultaneously integrate all three aspects, which limits classification performance. To address this challenge, this article proposes FS-CGNet, aiming to comprehensively exploit local, global, and frequency-domain information for more effective feature extraction and accurate characterization of different land cover types. First, FS-CGNet introduces the FDSFM, which utilizes the FFT to extract multiscale and multifrequency information from HSIs. This enhances the model's ability to capture both local structural features and global patterns while effectively reducing data redundancy. Next, the MSFAM is designed to integrate spectral information across different scales. By dynamically emphasizing key spectral channels through an attention mechanism, the model can focus on the most discriminative channels, further improving classification performance. Finally, the CGAM aggregates features from multiple spatial and spectral scales, ensuring that the model captures both local details and global structural information for a more sufficient feature representation.

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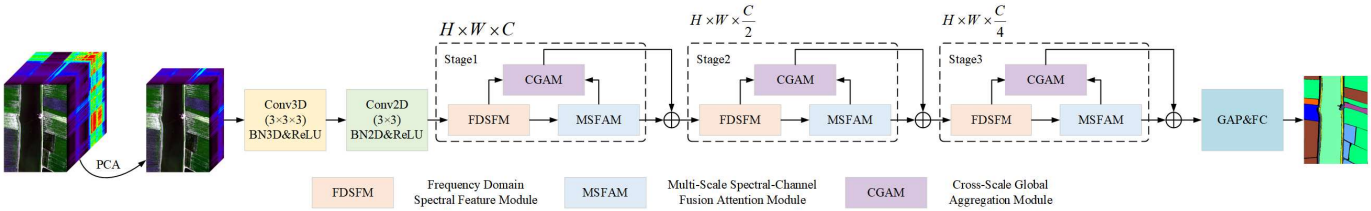


Fig. 1. Framework of the proposed FS-CGNet.

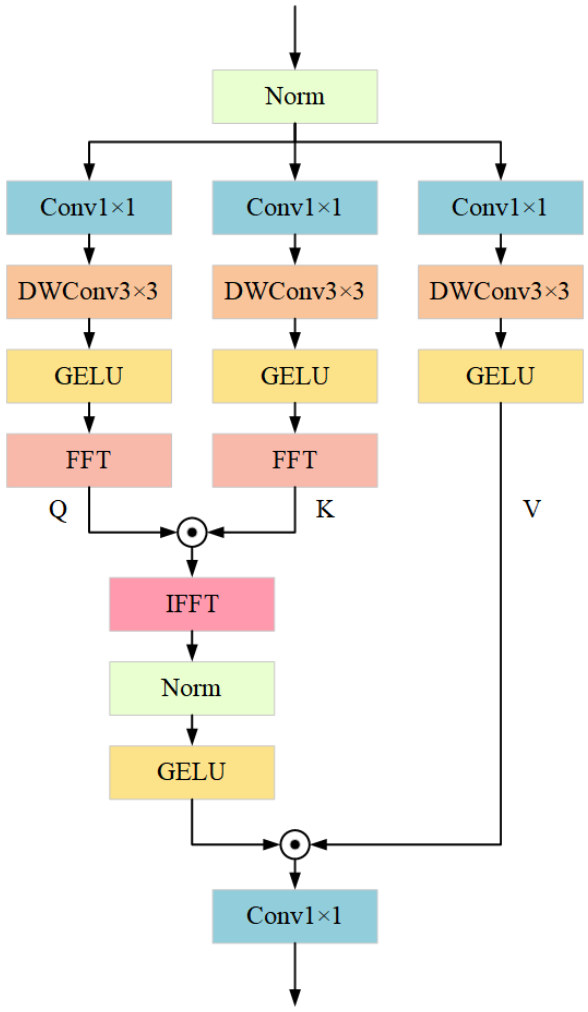


Fig. 2. Framework of the proposed FDSFM, where ⊙ represents the element-wise multiplication.

A. Frequency-Domain Spectral Feature Extraction Module

The architecture of the proposed FDSFM is illustrated in Fig. 2. Assume the input feature graph $X \in \mathbb{R}^{B\times C\times H\times W}$, where B represents the batch size; C is the number of channels; and H and W are the height and width of the feature graph, respectively. The module first processes the input feature map through a 1×1 convolution layer Conv2D

$$F = \text{Conv2D}_{1\times 1}(X) \in \mathbb{R}^{B\times(C\cdot 6)\times H\times W}. \tag{1}$$

Next, the module processes the F feature graph using a deeply separable convolution DWConv (i.e., convolution on each channel). The effect of this convolution is to convolve each channel and keep each channel independent. The convolution operation is as follows:

$$F_{dw} = \text{DWConv2D}_{3\times 3}(F) \in \mathbb{R}^{B\times(C\cdot 6)\times H\times W}. \tag{2}$$

Then, after applying the activation function GELU, the result is

$$F_{\text{GELU}} = \text{GELU}(F_{dw}). \tag{3}$$

Before processing in the frequency domain, the module rearranges the F_{GELU} feature map by patch size (in this case 9) so that it becomes a structure composed of multiple small blocks (patches). Suppose that the size of each small block is 9×9, and the rearranged shape is

$$Q, K, V = F_{\text{GELU}}.\text{chunk}(3, \text{dim} = 1). \tag{4}$$

Next, Fourier transforms Q and K, converting the features of the spatial domain to the frequency domain, and calculating their 2-D FFT, respectively,

$$Q_{\text{fft}} = \mathcal{F}\left(Q_{\text{patch}}\right) \tag{5}$$

$$K_{\text{fft}} = \mathcal{F}\left(K_{\text{patch}}\right) \tag{6}$$

where $\mathcal{F}(\cdot)$ represents the 2-D FFT. The Fourier transforms Q_{patch} and K_{patch} are the corresponding frequency components in the frequency domain.

In the frequency domain, the module calculates the similarity of Q_{patch} and K_{patch} by multiplying them element by element. The calculation formula is as follows:

$$\text{Out}_{\text{fft}} = Q_{\text{fft}} \times K_{\text{fft}}. \tag{7}$$

Then, the frequency-domain information is converted back to the spatial domain by inverse Fourier transform

$$\text{Out} = \mathcal{F}^{-1}(\text{Out}_{\text{fft}}) \tag{8}$$

where $\mathcal{F}^{-1}(\cdot)$ denotes the inverse 2-D Fourier transform. The result of the inverse Fourier transform is a 9×9 feature map that can be restored to the spatial domain.

The output feature map after the inversion is rearranged to restore the dimensions of the original inputs Q and K. Then, LayerNorm is used to normalize it so that each channel of the feature graph has a similar distribution. After that, the GELU activation function is used for the nonlinear transformation

$$\text{Out} = \text{GELU}\left(\text{LayerNorm}(\text{Out})\right). \tag{9}$$

Finally, multiply V and the processed Out element by element to get the final output feature plot, and restore the number of output channels to the original C through a 1×1 convolution layer

$$F_{\text{FDSFM}} = \text{Conv2D}_{1\times 1}(V \times \text{Out}) \in \mathbb{R}^{B\times C\times H\times W}. \tag{10}$$

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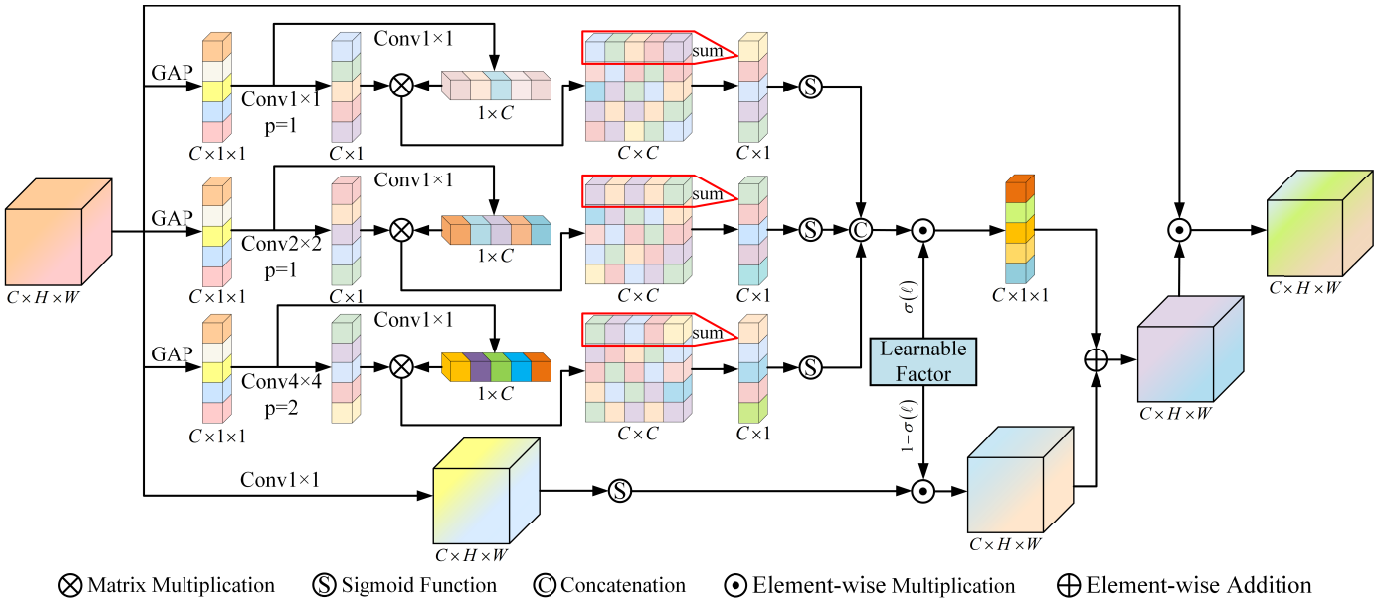


Fig. 3. Framework of the proposed MSFAM.

B. Multiscale Spectral–Channel Fusion Attention Module

The architecture of the proposed MSFAM is illustrated in Fig. 3. This module enhances the representation of input image features by calculating multiscale channel attention and spatial attention. Channel attention is computed using multiple 1-D convolution operations with different kernel sizes. The specific steps are as follows. First, the input is globally average pooled to obtain a feature map of size $[B, C, 1, 1]$, where B is the batch size and C is the number of channels. The response of each scale is then obtained by applying 1-D convolutions with different kernel sizes $s_1, s_2, \ldots, s_k$ to the pooled feature map. Afterward, the channel dimension is transformed by $\mathcal{M}_{fc}$ to obtain the channel attention for each scale.

The channel attention for each scale can be expressed by the following formula:

$$X_{\mathrm{CA}_i} = \sigma\left(\sum\left(\mathrm{Conv}_i(x) \cdot \mathcal{M}_{fc}(x)\right)\right) \tag{11}$$

where $\mathrm{Conv}_i(\cdot)$ represents the convolution operation of the i scale; x is the input after averaging pooling; $\mathcal{M}_{fc}(\cdot)$ is the channel convolution layer, used to adjust the channel dimension; and σ is the sigmoid activation function.

Finally, the channel attention of all scales is averaged to obtain multiscale channel attention

$$F_{\mathrm{MSA}} = \frac{1}{k}\sum_{i=1}^{k} X_{\mathrm{CA}_i}. \tag{12}$$

Spatial attention is extracted in the spatial dimension through convolution operations. Given input feature graph X, spatial attention is calculated as follows:

$$F_{\mathrm{SA}} = \sigma\left(\mathcal{M}_{fs}(X)\right) \tag{13}$$

where $\mathcal{M}_{fs}(\cdot)$ is a 1×1 convolution operation used to generate a spatial attention diagram.

A hybrid module, Mix, is defined, which generates the final output by weighted fusion of two input feature graphs F_1 and F_2. Specifically, the input of Mix module is two feature maps, the weight is controlled by a learnable parameter ℓ, and the fusion mode uses the sigmoid activation function to adjust the fusion ratio.

Given input feature plots F_1 and F_2, the fusion process can be represented by the following formula:

$$F_{\mathrm{Mix}} = F_1 \cdot \sigma(\ell) + F_2 \cdot (1 - \sigma(\ell)) \tag{14}$$

where $\sigma(\ell)$ is the sigmoid activation function: $\sigma(\ell) = 1/(1 + e^{-\ell})$ and F_{Mix} is the output feature map.

The multiscale channel attention and spatial attention are weighted by the Mix module. The final fusion output F_{MSFAM} can be expressed as

$$F_{\mathrm{MSFAM}} = X \cdot \mathrm{Mix}\left(F_{\mathrm{MSA}}, F_{\mathrm{SA}}\right). \tag{15}$$

C. Cross-Scale Global Aggregation Module

The architecture of the proposed CGAM is illustrated in Fig. 4. Suppose that the two input feature graphs are L (left feature graph) and R (right feature graph), both of which have the shape $B \times C \times H \times W$. First, make sure that the L and R feature plots have the same spatial dimensions. If their spatial dimensions are different, the size of the R feature map is adjusted to the size of L by the upsampling operation

$$\hat{R} = \mathrm{upsample}(R) \quad \text{if} \quad \mathrm{shape}(L) \neq \mathrm{shape}(R) \tag{16}$$

where $\hat{R}$ is the R feature map after upsampling so that it has the same size as the L feature map.

After dimensional alignment, CGAM performs feature fusion on the input L and the adjusted R. First, the convolution operation $\mathrm{Conv}_{\mathrm{CBR}}$ is performed on the R feature graph, and then, the convolutions R and L are concatenated to form a new fusion feature graph. At this time, the convolution operation $\mathrm{Conv}_{\mathrm{cat}}$ is used to further process the spliced feature map and generate the fused feature map O

$$O = \mathrm{Conv}_{\mathrm{cat}}\left(L, \hat{R}_1\right) \tag{17}$$

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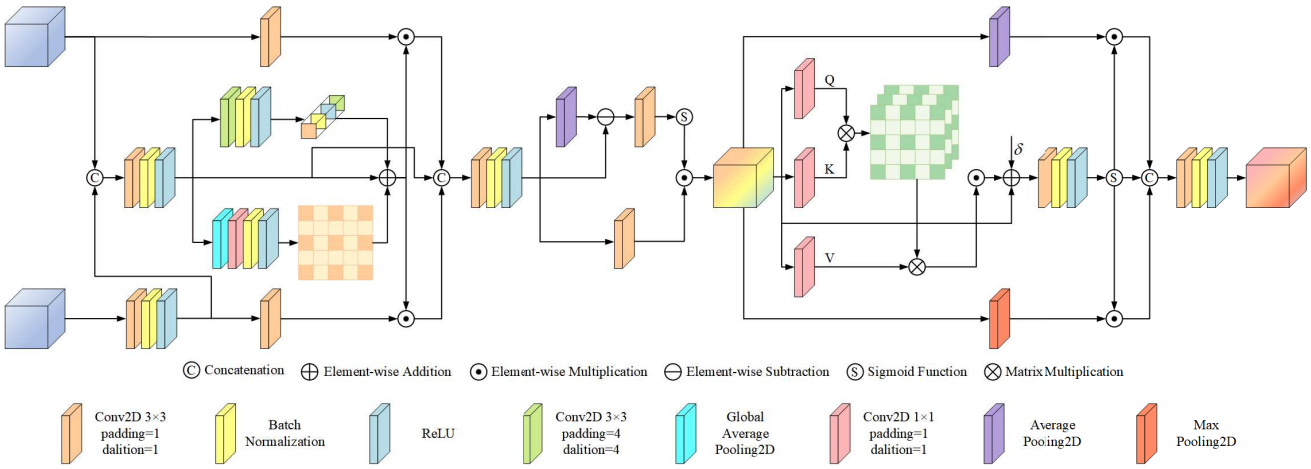


Fig. 4. Framework of the proposed CGAM.

where L is the original feature map and $\hat{R}_1 = \mathrm{Conv}_{\mathrm{CBR}}(R)$, which represents the feature map R after the convolution operation. $\mathrm{Conv}_{\mathrm{CBR}}(\cdot)$ refers to a sequence that includes a 3×3 convolution layer, a batch normalization layer, and a ReLU activation function.

In order to capture global context information, CGAM uses dilated convolution to expand the receptive field of feature graphs. Dilative convolution can increase the range of receptive fields, thus capturing a wider range of context information. After the expanded convolution operation is performed on the fused feature graph O, it is added with the original feature graph to generate the context-enhanced feature graph O_g

$$O_g = O + \mathrm{DilatedConv}(O). \tag{18}$$

Then, the global average pooling (GAP) operation is used to process the feature map O and extract the global information. The global features are compressed by 1×1 convolution, and the scaled global features are obtained. Then, bilinear interpolation is used to sample the global feature map to the spatial size of the original input feature map

$$O_g = O_g + \mathrm{Upsample}\left(\mathrm{Conv2D}_{1\times1}(\mathrm{GAP}(O))\right). \tag{19}$$

After obtaining the context-enhanced feature graphs, CGAM performs convolution operations on the L and R feature graphs and weights them using the context-enhanced feature graphs O_g. The weighted L and R feature maps are spliced with the previous feature maps O and further fused into a new feature map O_2 by a convolution operation

$$O_2 = \mathrm{Conv}_{\mathrm{cat2}}\left(L_2, \hat{R}_2, O\right) \tag{20}$$

where $L_2 = \mathrm{Conv2D}(L) \odot O$ and $\hat{R}_2 = \mathrm{Conv2D}(\hat{R}_1) \odot O$ are the feature graphs after convolution and weighting.

Finally, CGAM generates the weighting coefficient through local mean pooling and local maximum pooling operations. The pooling result is subtracted from the local mean pooling result and the convolution operation is used to generate the weighting coefficient. The sigmoid activation function was used to normalize the weighting coefficient to the interval $[0, 1]$ and multiply it with the O_2 feature map element by element to obtain the final output feature map O_f

$$O_f = O_2 \otimes O_{g2} \tag{21}$$

where O_{g2} is the dynamic weighting coefficient obtained through the above weighting process and $\otimes$ denotes the matrix multiplication.

Three convolution layers with 1×1 kernels are applied to enhance the nonlinearity of the previously obtained feature map, resulting in the generation of three distinct feature matrices

$$\begin{cases} Q = \left(\mathrm{reshape}\left(\mathrm{Conv2D}_{1\times1}\left(O_f\right)\right)\right)^T \\ K = \left(\mathrm{reshape}\left(\mathrm{Conv2D}_{1\times1}\left(O_f\right)\right)\right) \\ V = \left(\mathrm{reshape}\left(\mathrm{Conv2D}_{1\times1}\left(O_f\right)\right)\right). \end{cases} \tag{22}$$

The similarity between Q and K is calculated (the weighted matrix is obtained by matrix multiplication) and the weighted matrix is multiplied by the matrix with V to obtain the weighted feature graph

$$G = V \otimes \mathrm{Softmax}(Q \otimes V) \tag{23}$$

where G represents the enhanced features obtained by adaptive weighting.

The enhanced feature G is added to the original feature graph O_f and then convolved and normalized by the sigmoid activation function

$$O_G = \sigma\left(\mathrm{Conv}_{\mathrm{CBR}}\left(\delta \otimes G + O_f\right)\right) \tag{24}$$

where δ is a learnable parameter that controls the weight of the enhanced feature.

Finally, average and maximum pooling operations are performed on the input features O_f, the results of the two are spliced together, and the final output is obtained through a convolutional layer

$$F_{\mathrm{CGAM}} = \mathrm{Conv}_{\mathrm{CBR}}\left(\mathrm{Conv}_{\mathrm{cat}}\left(\mathrm{Avg}\left(O_f\right), \mathrm{Max}\left(O_f\right)\right)\right). \tag{25}$$

The procedural steps of the proposed FS-CGNet are summarized in Algorithm 1.

III. Experiments and Discussion

In this section, we perform a comprehensive set of experiments to assess the performance and effectiveness of the FS-CGNet model. These experiments are conducted using four publicly available HSI datasets to thoroughly evaluate

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Algorithm 1 Pseudo Procedure of the Proposed FS-CGNet

Input: Input origin HSI data $X \in \mathbb{R}^{H\times W\times D}$, patch size B, the number of bands after PCA processing C.

Output: Predicted labels of the testing samples.

1: Apply PCA for dimension reduction $I_{PCA} \in \mathbb{R}^{H\times W\times C}$;
2: **FDSFM:**
3: Perform depthwise separable convolution on input feature map, followed by activation and feature splitting into Q, K, and V.
4: Compute similarity between Q and K in the frequency domain, apply inverse Fourier transform and GELU activation function, combine with V to obtain output F_{FDSFM}.
5: **MSFAM:**
6: Perform global average pooling on F_{FDSFM}, apply multi-scale channel attention using 1D convolutions, and average the results to obtain the multi-scale channel attention F_{MSA}.
7: Calculate spatial attention $\mathcal{M}_{fs}(X)$ with Sigmoid activation function to obtain the spatial attention F_{SA}.
8: Fuse F_{MSA} and F_{SA} with a learnable parameter ℓ, and multiply by input F_{FDSFM} to obtain output F_{MSFAM}.
9: **CGAM:**
10: Align left feature map L and the right feature map R, then fuse them using convolution.
11: Enhance context with dilated convolution and global average pooling.
12: Apply dynamic weighting, followed by Softmax refinement.
13: Fuse features using the Sigmoid activation function and pooling operation, generating the final output F_{CGAM}.
14: Use the trained model to obtain the test labels.

the model's capabilities and accuracy in various scenarios. To provide a comprehensive comparison, we choose nine DL models for benchmarking: SSFTT [48], morphFormer [52], DBCT [50], GAHT [53], MASSFormer [51], DCTN [59], LSFAT [49], MambaHSI [15], and 3DSS-Mamba [56]. In order to evaluate and compare the performance of the different models, we employ three standard evaluation metrics: overall accuracy (OA), average accuracy (AA), and the kappa coefficient (κ). These metrics will help us perform a thorough and well-rounded analysis of each model's effectiveness in the classification of HSIs. Additionally, all of the experiments were carried out on a high-performance computing setup, which includes an Intel Core i7-13700KF processor, a powerful NVIDIA GeForce RTX 4090 GPU with 24 GB of dedicated memory, and 64 GB of RAM. This configuration ensured smooth and efficient execution while providing the necessary computational power to handle the resource-intensive nature of the experiments.

A. Dataset Description

For our experiments, we utilized four well-known and publicly accessible HSI datasets, each commonly used in the HSI classification domain. These datasets include the Pavia University dataset, Salinas dataset, WHU-Hi-LongKou dataset, and the Xiong An dataset [60]. These datasets are widely acknowledged within the research community due to their diverse and challenging nature, providing various test scenarios for evaluating model performance. Their wide range of complexity in data and environmental conditions allows for a thorough evaluation of how effectively the model can process and classify different types of hyperspectral data.

The first dataset, sourced from the University of Pavia, was collected by the Reflective Optical Systems Imaging Spectrometer (ROSIS) on July 8, 2002, over the city of Pavia in northern Italy. It consists of 115 spectral bands and has spatial dimensions of 610 $\times$ 340 pixels. The spectral resolution is 4 nm, covering the range from 0.43 to 0.86 μm, while the spatial resolution is 3.7 m/pixel. The ground truth (GT) for this dataset includes nine distinct urban land cover classes, with a total of 42 776 annotated pixels. After eliminating 12 noisy bands, the remaining 103 bands were employed for classification purposes.

The Salinas dataset was collected from the Salinas Valley, located in California, USA, in 1998 using the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor. This dataset focuses on an agricultural area within the Salinas Valley and consists of an image with spatial dimensions of 512 $\times$ 217 pixels, along with 224 spectral bands. The spatial resolution of each spectral band is 3.7 m/pixel. For the purpose of this analysis, 20 spectral bands related to water absorption were removed, leaving a total of 204 bands for further examination. The dataset includes 16 GT classes, representing a variety of land cover types, such as agricultural fields with vegetables, vineyard areas, and bare soils, totaling 54 129 pixels.

The WHU-Hi-LongKou dataset was recorded on July 17, 2018, between 13:49 and 14:37, in Longkou Town, Hubei Province, China, using an 8-mm focal length Headwall Nano-Hyperspec imaging sensor mounted on a DJI Matrice 600 Pro (DJI M600 Pro) UAV platform. The data acquisition was carried out under clear skies and cloud-free conditions, with the temperature around 36 °C and a relative humidity of about 65%. The region of interest consists of a typical agricultural landscape, where six different crop species are cultivated: corn, cotton, sesame, broad-leaf soybean, narrow-leaf soybean, and rice. The UAV operated at an altitude of 500 m, capturing images with a resolution of 550 $\times$ 400 pixels, spanning a total of 270 spectral bands, ranging from 400 to 1000 nm. The spatial resolution of the HSIs captured by the UAV is approximately 0.463 m/pixel.

The hyperspectral imaging data for Ma Tiwan Village, located in the Xiong'an New Area, were obtained using the full-spectrum multimodal imaging spectrometer developed by Shanghai Institute of Technical Physics, part of Chinese Academy of Sciences, as a key component of the High-Resolution Earth Observation Program. This advanced spectrometer spans a spectral range from 400 to 1000 nm, encompassing 250 individual spectral bands. The captured imagery has dimensions of 3750 $\times$ 1580 pixels and boasts a spatial resolution of 0.5 m, ensuring exceptional accuracy for conducting in-depth remote sensing analysis.

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TABLE I
DISTRIBUTION OF TRAINING AND TESTING SAMPLES FOR EACH CLASS IN THE PAVIA UNIVERSITY DATASET WITH CORRESPONDING SPECTRAL CLASS COLORS

NO.	Color	Class	Train	Test	Total
1		Asphalt	663	5968	6631
2		Meadows	1865	16784	18649
3		Gravel	210	1889	2099
4		Trees	306	2758	3064
5		Painted Metal Sheets	134	1211	1345
6		Bare Soil	503	4526	5029
7		Bitumen	133	1197	1330
8		Self-Blocking Bricks	368	3314	3682
9		Shadows	95	852	947
-	-	Total	4277	38499	42776

TABLE II
DISTRIBUTION OF TRAINING AND TESTING SAMPLES FOR EACH CLASS IN THE SALINAS DATASET WITH CORRESPONDING SPECTRAL CLASS COLORS

NO.	Color	Class	Train	Test	Total
1		Brocoli-Green-Weeds1	201	1808	2009
2		Brocoli-Green-Weeds2	372	3354	3726
3		Fallow	197	1779	1976
4		Fallow-Rough-Plow	139	1255	1394
5		Fallow-Smooth	268	2410	2678
6		Stubble	396	3563	3959
7		Celery	358	3221	3579
8		Grapes-Untrained	1127	10144	11271
9		Soil-Vinyard-Develop	620	5583	6203
10		Corn-Senesced-Green-Weeds	328	2950	3278
11		Leetuce-Romaine-4wk	107	961	1068
12		Leetuce-Romaine-5wk	193	1734	1927
13		Leetuce-Romaine-6wk	91	825	916
14		Leetuce-Romaine-7wk	107	963	1070
15		Vinyard-Untrained	727	6541	7268
16		Vinyard-Vertical-trellis	181	1626	1807
-	-	Total	5412	48717	54129

TABLE III
DISTRIBUTION OF TRAINING AND TESTING SAMPLES FOR EACH CLASS IN THE WHU-HI-LONGKOU DATASET WITH CORRESPONDING SPECTRAL CLASS COLORS

NO.	Color	Class	Train	Test	Total
1		Corn	3451	31060	34511
2		Cotton	837	7537	8374
3		Sesame	303	2728	3031
4		Broad-Leaf Soybean	6321	56891	63212
5		Narrow-Leaf Soybean	415	3736	4151
6		Rice	1185	10669	11854
7		Water	6706	60350	67056
8		Roads and Houses	712	6412	7124
9		Mixed Weed	523	4706	5229
-	-	Total	20453	184089	204542

TABLE IV
DISTRIBUTION OF TRAINING AND TESTING SAMPLES FOR EACH CLASS IN THE XIONG AN DATASET WITH CORRESPONDING SPECTRAL CLASS COLORS

NO.	Color	Class	Train	Test	Total
1		Water	22565	203082	225647
2		Rice	18077	162689	180766
3		Acer Compound	1535	13818	15353
4		Grass Land	45214	406930	452144
5		Wilow	47559	428032	475591
6		Japonica Sophra	16934	152408	169342
7		White Wax	2330	20974	23304
8		Corn	16565	149082	165647
9		Peach	3841	34568	38409
10		Pear Tree	19383	174447	193830
11		Poplar	561	5051	5612
12		Soybean	5917	53248	59165
13		Vegetable Fields	102651	923862	1026513
14		Elm	715	6436	7151
15		Bare Land	9107	81965	91072
16		Koelreuteria Paniclata	2915	26233	29148
17		Rice Stubble	150	1346	1496
18		Locust	42179	379611	421790
19		Spatse Forest	6551	58963	65514
20		Houses	2962	26654	29616
-	-	Total	367711	3309399	3677110

In this study, a random selection approach was used to choose 10% of the total sample size from each dataset: Pavia University, Salinas, WHU-Hi-LongKou, and Xiong An, to be used as training samples. The remaining 90% of the samples from each dataset were reserved for the testing phase. The detailed information on the land cover categories, as well as the number of samples allocated for training and testing, is carefully summarized in Tables I–IV. Additionally, to provide a clearer understanding of the datasets, Figs. 5–8 present the false-color images for each dataset. These images display the GT maps and highlight the corresponding color codes used to differentiate the various land cover categories within the datasets.

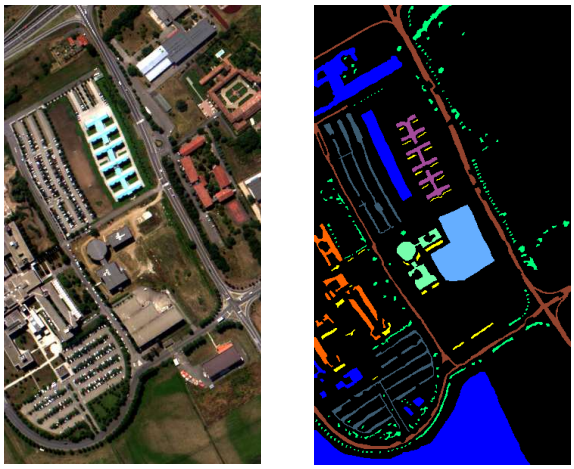
B. Experimental Result

Table V shows the evaluation results of different methods on the Pavia University dataset, including class accuracy (%), OA, AA, and κ. From the table, it is evident that FS-CGNet performs exceptionally well across all evaluation metrics, particularly in the κ coefficient, which reached 99.77%, surpassing all other methods. This demonstrates a high level of consistency between the classification results and the true GT, indicating that the algorithm has excellent reliability and stability. Compared to other algorithms, FS-CGNet stands out

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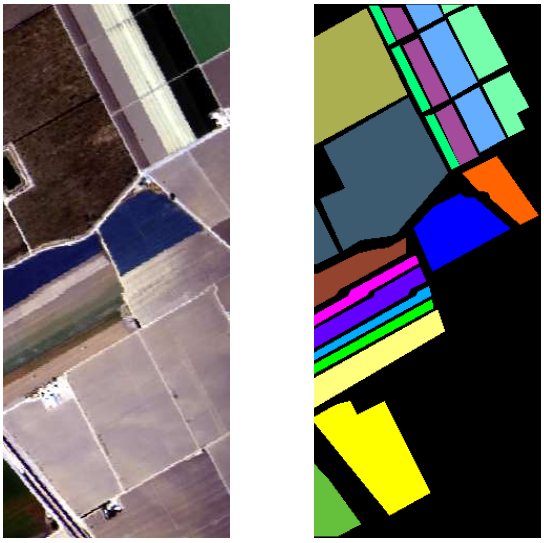
TABLE V
EVALUATION OF DIFFERENT METHODS ON THE PAVIA UNIVERSITY DATASET BASED ON CLASS ACCURACY (%), OA (%), AA (%), AND κ (%)

NO.	SSFTT	morphFormer	DBCT	GAHT	MASSFormer	DCTN	LSFAT	MambaHSI	3DSS-Mamba	FS-CGNet
1	99.77	99.09	99.62	99.04	99.27	99.42	99.55	98.21	99.68	**99.85**
2	99.91	99.90	99.93	99.91	99.87	**99.98**	**99.98**	99.85	99.87	99.95
3	99.42	96.68	**99.84**	99.46	96.69	99.08	98.42	96.57	97.62	99.52
4	96.24	99.34	99.09	98.16	99.20	98.45	98.27	95.47	**99.45**	98.95
5	**100.0**	99.67	99.92	99.83	**100.0**	**100.0**	94.17	88.27	**100.0**	99.92
6	99.87	97.00	**100.0**	99.98	97.69	99.98	99.80	99.89	99.54	99.98
7	99.01	97.28	**100.0**	99.01	98.56	**100.0**	98.43	97.63	**100.0**	99.92
8	98.38	90.57	97.78	98.10	88.58	94.66	94.36	90.26	88.05	**99.82**
9	97.21	98.81	99.64	97.73	**100.0**	**100.0**	**100.0**	99.31	99.88	99.63
OA (%)	99.38	98.27	99.63	99.40	98.23	99.26	98.95	97.76	98.54	**99.83**
AA (%)	98.87	97.59	99.53	99.02	97.76	99.06	98.11	96.16	98.23	**99.73**
κ (%)	99.18	97.71	99.51	99.21	97.66	99.03	98.61	97.02	98.07	**99.77**



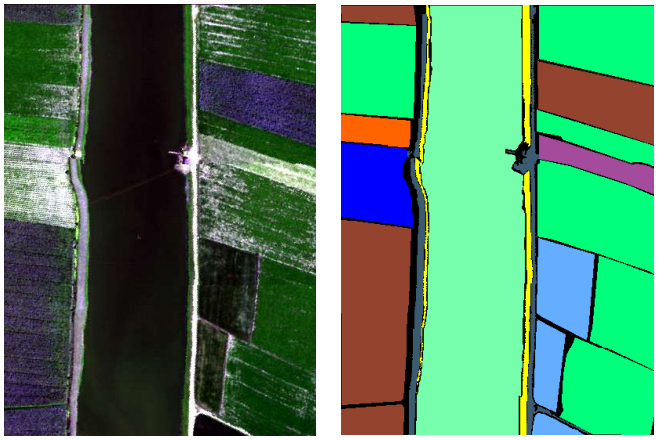
(a) (b)

Fig. 5. Pavia University dataset. (a) Three-band color composite image. (b) GT map.



(a) (b)

Fig. 6. Salinas dataset. (a) Three-band color composite image. (b) GT map.



(a) (b)

Fig. 7. WHU-Hi-LongKou dataset. (a) Three-band color composite image. (b) GT map.



(a) (b)

Fig. 8. Xiong An dataset. (a) Three-band color composite image. (b) GT map.

overall. Although methods like DBCT and GAHT also perform well, FS-CGNet exhibits higher accuracy across multiple classes, particularly in complex spectral features and categories with fewer training samples. FS-CGNet significantly

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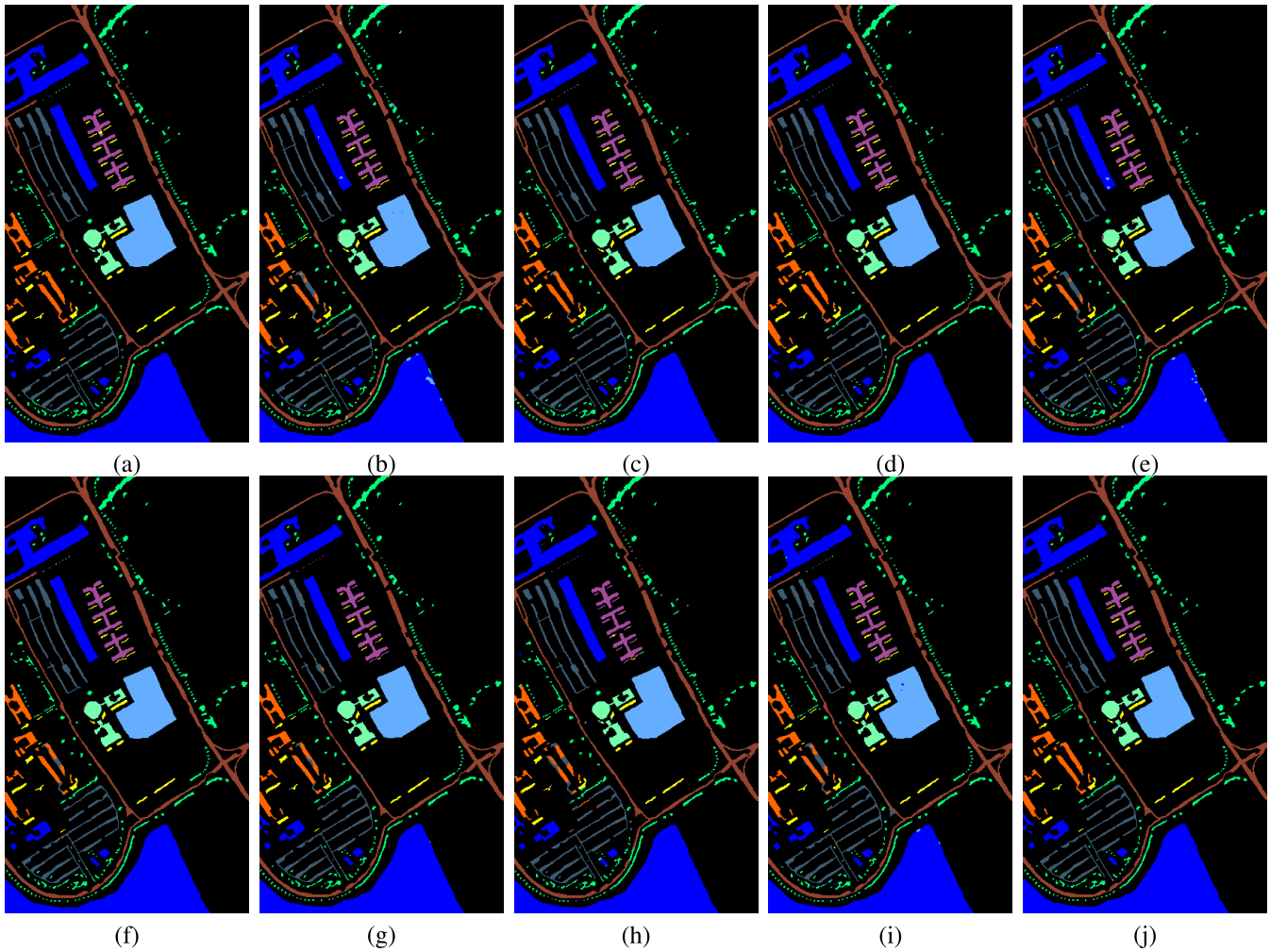


Fig. 9. Comparison of classification results on the Pavia University dataset using various methods with only 10% of the training samples. (a) SSFTT (OA = 99.38%). (b) morphFormer (OA = 98.27%). (c) DBCT (OA = 99.63%). (d) GAHT (OA = 99.40%). (e) MASSFormer (OA = 98.23%). (f) DCTN (OA = 99.26%). (g) LSFAT (OA = 98.95%). (h) MambaHSI (OA = 97.76%). (i) 3DSS-Mamba (OA = 98.54%). (j) FS-CGNet (OA = 99.83%).

improves classification accuracy. Its high κ, OA, and AA further confirm its outstanding performance in ground object classification tasks. Fig. 9 provides a visual comparison of the GT and classification maps for each algorithm, further showcasing the advantages of FS-CGNet in ground object classification. Through image comparison, it is clear that FS-CGNet achieves more precise classification in complex ground object categories. Compared to other algorithms, the classification results are more detailed and accurate. This further verifies that FS-CGNet can effectively improve classification stability and reliability when handling datasets with complex spectral features.

Table VI presents the classification results of ten algorithms, including FS-CGNet, on the Salinas dataset, while Fig. 10 shows the GT and classification maps for each algorithm. Among the ten algorithms, FS-CGNet stands out with the highest overall performance across all evaluated metrics, achieving an OA of 99.94%, an AA of 99.90%, and a κ of 99.94%. These results indicate FS-CGNet's exceptional ability to accurately classify the Salinas dataset with great precision and stability. Other methods, such as SSFTT and morphFormer, also demonstrate strong performance, with OAs of 99.38% and 98.27%. While these algorithms perform well, FS-CGNet still maintains a clear advantage in classification accuracy, particularly when dealing with more complex or varied classes. For example, FS-CGNet excels in distinguishing difficult-to-classify categories, showing a more robust ability to maintain high accuracy under diverse conditions. Although algorithms such as MASSFormer, MambaHSI, and 3DSS-Mamba perform well in overall classification, they show slight performance degradation in the classification accuracy of specific categories. For example, in the classification of category 8, MASSFormer, MambaHSI, and 3DSS-Mamba slightly underperform compared to FS-CGNet, achieving category accuracies of 94.54%, 96.54%, and 97.56%, while FS-CGNet achieved 100% accuracy. Similarly, the classification results for category 15 also show some disparity, with FS-CGNet leading with 99.83% accuracy, while the performance of other algorithms slightly declined. This indicates that FS-CGNet has an advantage when handling categories with complex backgrounds or details, maintaining higher classification accuracy. Overall, the findings suggest that FS-CGNet is the most

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TABLE VI
EVALUATION OF DIFFERENT METHODS ON THE SALINAS DATASET BASED ON CLASS ACCURACY (%), OA (%), AA (%), AND κ (%)

NO.	SSFTT	morphFormer	DBCT	GAHT	MASSFormer	DCTN	LSFAT	MambaHSI	3DSS-Mamba	FS-CGNet
1	99.94	**100.0**	95.81	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**
2	**100.0**	99.91	**100.0**	99.03	**100.0**	**100.0**	**100.0**	99.82	**100.0**	**100.0**
3	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**
4	99.92	99.29	**100.0**	99.20	98.74	97.88	99.52	96.60	99.44	98.74
5	97.72	99.67	99.79	99.09	99.96	99.71	99.46	98.54	99.75	**100.0**
6	99.89	**100.0**	**100.0**	99.89	**100.0**	98.51	99.86	99.86	**100.0**	**100.0**
7	99.91	**100.0**	**100.0**	**100.0**	**100.0**	99.97	**100.0**	**100.0**	**100.0**	**100.0**
8	99.45	97.50	98.02	98.39	94.54	98.12	99.35	96.54	97.56	**100.0**
9	99.95	99.91	99.71	99.70	99.64	99.61	99.93	99.95	99.73	**100.0**
10	99.97	98.98	99.01	98.88	99.55	99.56	99.69	99.53	99.76	**100.0**
11	**100.0**	97.46	96.68	98.56	97.85	**100.0**	99.79	99.07	99.38	99.90
12	99.71	99.83	98.92	88.69	99.71	85.84	97.42	98.92	99.88	**100.0**
13	98.33	99.88	**100.0**	**100.0**	**100.0**	**100.0**	97.46	99.14	**100.0**	**100.0**
14	80.17	99.58	**100.0**	92.98	98.96	86.96	99.47	99.68	98.97	**100.0**
15	95.42	93.37	97.16	96.60	97.15	97.62	98.51	91.55	95.60	**99.83**
16	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**
OA (%)	98.59	98.39	98.83	98.34	98.28	98.16	99.43	97.83	98.79	**99.94**
AA (%)	98.15	99.09	99.07	98.19	99.13	97.74	99.40	98.70	99.38	**99.90**
κ (%)	98.43	98.21	98.70	98.15	98.09	97.95	99.37	97.58	98.65	**99.94**

TABLE VII
EVALUATION OF DIFFERENT METHODS ON THE WHU-HI-LONGKOU DATASET BASED ON CLASS ACCURACY (%), OA (%), AA (%), AND κ (%)

NO.	SSFTT	morphFormer	DBCT	GAHT	MASSFormer	DCTN	LSFAT	MambaHSI	3DSS-Mamba	FS-CGNet
1	99.85	99.40	99.90	99.84	99.55	99.90	99.81	99.33	99.60	**99.98**
2	99.73	97.17	99.56	99.67	98.20	99.31	94.93	96.89	96.12	**99.91**
3	99.52	99.29	**100.0**	99.85	97.93	**100.0**	**100.0**	99.91	99.31	**100.0**
4	99.46	98.87	98.88	99.42	99.21	96.78	98.68	98.12	97.85	**99.98**
5	99.52	96.45	99.79	**99.88**	97.02	99.64	96.75	99.29	96.60	99.65
6	97.29	99.65	99.96	99.33	99.33	99.39	99.15	97.85	99.54	**99.98**
7	99.23	99.96	99.93	99.72	99.88	99.78	99.83	99.08	99.96	**99.99**
8	90.17	95.81	95.42	86.48	97.47	83.55	92.48	74.32	98.33	**98.73**
9	93.19	93.52	97.49	95.82	96.89	**99.73**	97.55	94.76	93.83	99.25
OA (%)	98.88	99.01	99.36	99.03	99.27	98.14	98.85	97.55	98.78	**99.91**
AA (%)	97.55	97.79	98.99	97.78	98.39	97.57	97.69	95.50	97.90	**99.72**
κ (%)	98.52	98.70	99.16	98.73	99.04	97.54	98.49	96.77	98.39	**99.88**

effective algorithm for the classification of the Salinas dataset, consistently outperforming the other methods in terms of accuracy, robustness, and stability.

Table VII lists the classification performance of ten different algorithms on the WHU-Hi-LongKou dataset, specifically including the classification accuracy of each algorithm on this dataset. Additionally, Fig. 11 presents the classification results of each algorithm. From the table, it is clear that FS-CGNet outperforms all other methods across all evaluation metrics. Specifically, FS-CGNet achieves the highest OA of 99.94%, AA of 99.90%, and κ of 99.94%. Compared to this, other algorithms like DBCT and LSFAT also perform excellently in terms of OA, reaching close to or exceeding 99%, but they still fall short in AA and κ compared to FS-CGNet. For example, LSFAT achieves an OA of 99.43% but has an AA of 99.40%, which is lower than FS-CGNet. The results demonstrate FS-CGNet's superior ability to achieve high performance across all aspects of classification. In summary, FS-CGNet emerges as the top performer on the WHU-Hi-LongKou dataset, demonstrating superior classification accuracy across multiple evaluation metrics. Other algorithms like DBCT and DCTN show solid results but fall short in terms of consistency, especially in categories that require fine spectral distinction.

Table VIII presents the classification performance of ten different algorithms applied to the Xiong An hyperspectral dataset. The table provides a detailed comparison of the accuracy rates achieved by each method. Meanwhile, Fig. 12 visually represents the classification outcomes for each algorithm, offering a graphical depiction of their performance. As shown in the table, FS-CGNet outperforms all other algorithms across all metrics, achieving the highest OA of 99.98%, AA of 99.97%, and κ of 99.98%. FS-CGNet consistently delivers superior performance, with OA values reaching 100% for individual test cases, while other algorithms such as SSFTT, morphFormer, and DBCT achieve high but slightly lower OA values, with 99.19%, 98.70%, and 98.91%, respectively. For AA, FS-CGNet again leads with 99.97%, far surpassing other methods like SSFTT with 98.29%, and morphFormer with 95.84%. Similarly, in terms of κ, FS-CGNet achieves 99.98%, which is higher than other algorithms such as SSFTT (99.05%) and morphFormer (98.49%). Although methods like GAHT, MambaHSI, and DCTN show good performance with

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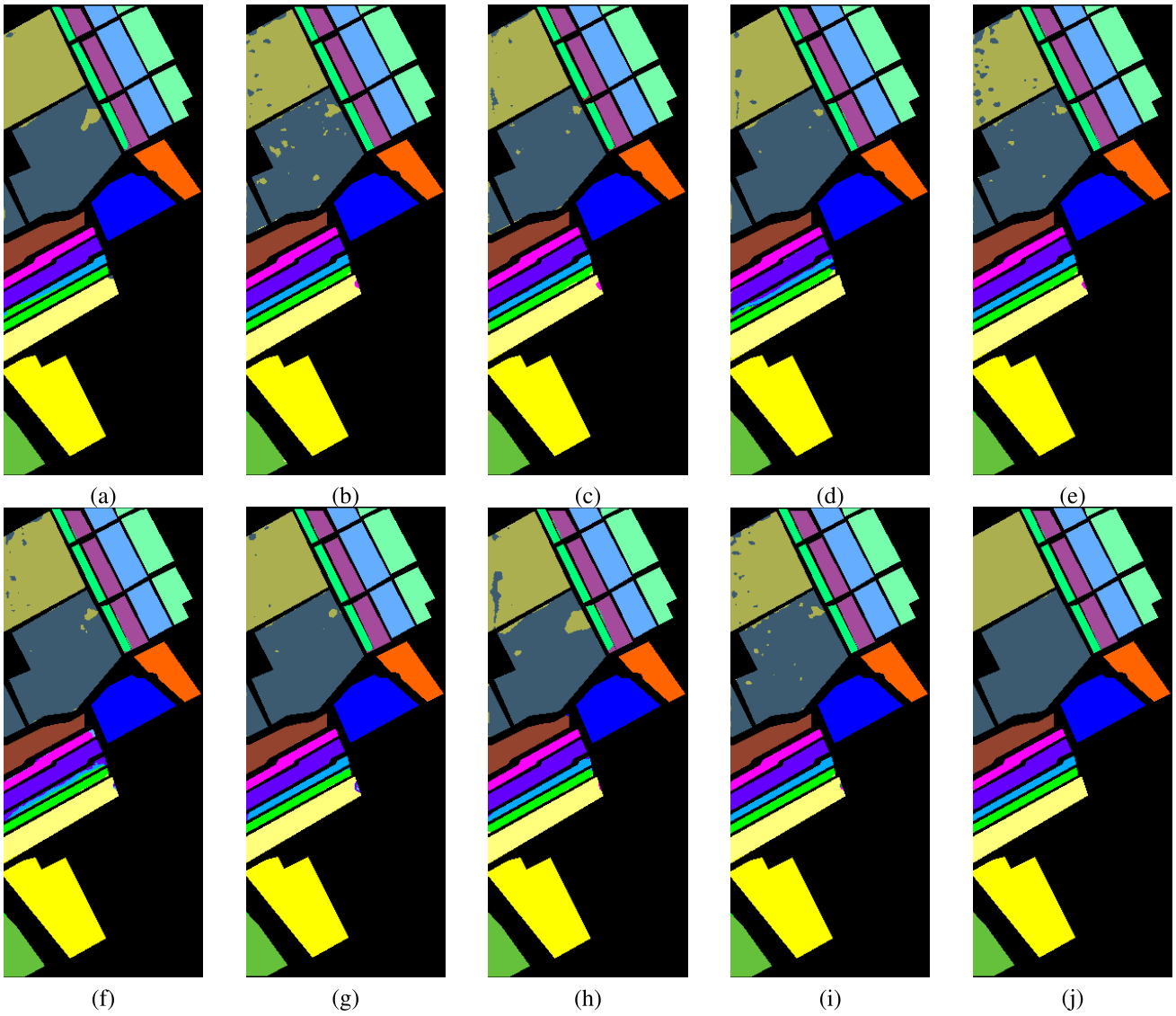


Fig. 10. Comparison of classification results on the Salinas dataset using various methods with only 10% of the training samples. (a) SSFTT (OA = 98.59%). (b) morphFormer (OA = 98.39%). (c) DBCT (OA = 98.83%). (d) GAHT (OA = 98.34%). (e) MASSFormer (OA = 98.28%). (f) DCTN (OA = 98.16%). (g) LSFAT (OA = 99.43%). (h) MambaHSI (OA = 97.83%). (i) 3DSS-Mamba (OA = 98.79%). (j) FS-CGNet (OA = 99.94%).

OA values above 97%, FS-CGNet still stands out in all metrics, consistently achieving perfect scores in most test cases. These results demonstrate that FS-CGNet excels in handling various challenges, ensuring robust and consistent classification performance.

Fig. 13 shows the confusion matrix for the HSI classification of the proposed FS-CGNet model on the Pavia University dataset. According to the analysis of the confusion matrix, the classification performance of the proposed FS-CGNet model is generally strong, with low misclassification rates for most categories. Specifically, misclassifications are more significant for categories 1–4, 8, and 9. For example, in category 1, 3 pixels are misclassified as category 2, and 6 pixels are misclassified as category 4; in category 4, 2 pixels are misclassified as category 1, 5 pixels are misclassified as category 2, 2 pixels are misclassified as category 3, 13 pixels are misclassified as category 8, and 7 pixels are misclassified as category 9. In contrast, misclassifications for categories 5–7 are relatively few, with only one misclassified pixel. The reasons for these misclassifications may include three aspects. First, noise interference, such as atmospheric conditions or sensor issues, may introduce noise during data collection, affecting the accuracy of land cover identification. Second, the "same object, different spectra" phenomenon, where the same land cover class exhibits different spectral characteristics across different spectral channels. Third, the "same spectrum, different objects" phenomenon, where different land cover classes exhibit similar spectral characteristics in the same spectral channel.

C. Complexity Analysis

The FS-CGNet model consists of three main modules: FDSFM, MSFAM, and CGAM, each with different computational complexities. The FDSFM has a computational complexity of $O(L \log L)$, where L is the size of the input

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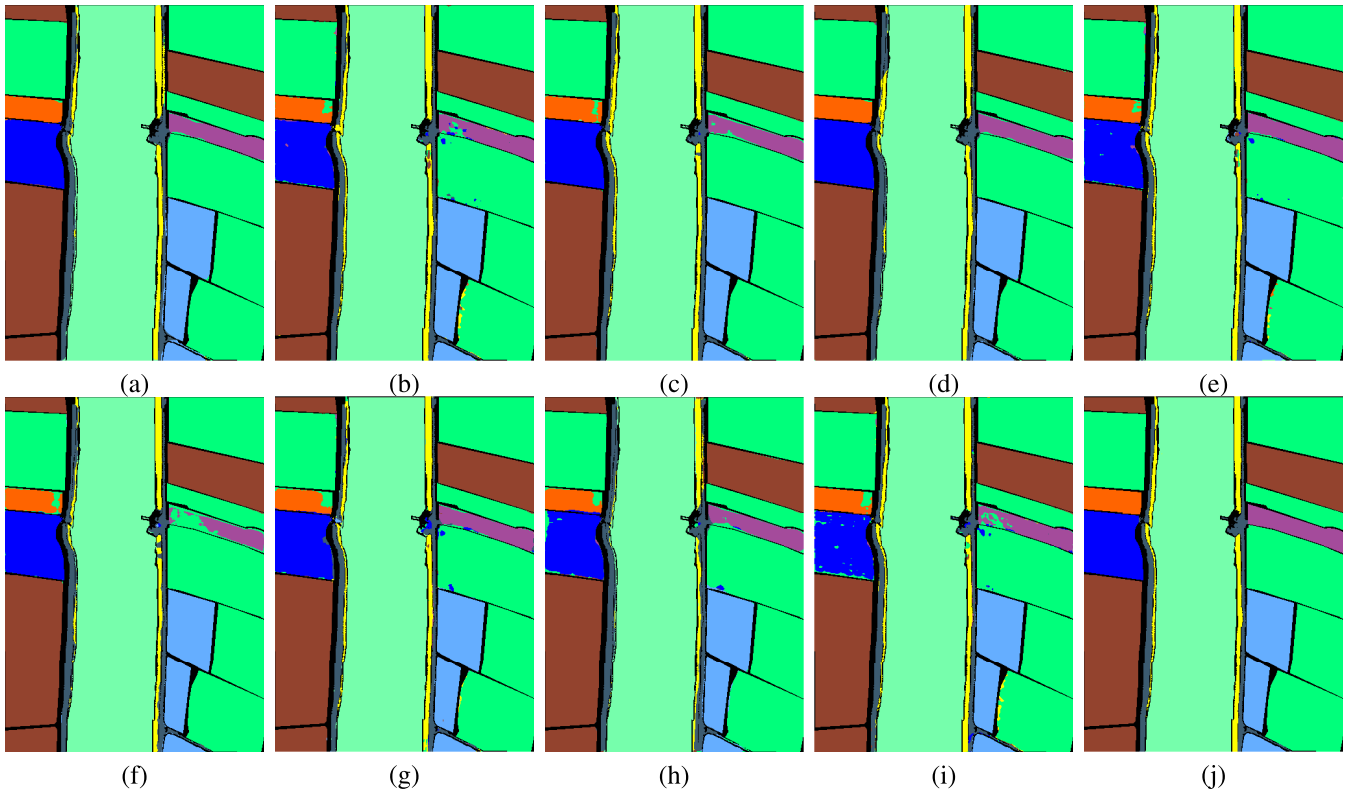


Fig. 11. Comparison of classification results on the WHU-Hi-LongKou dataset using various methods with only 10% of the training samples. (a) SSFTT (OA = 98.88%). (b) morphFormer (OA = 99.01%). (c) DBCT (OA = 99.36%). (d) GAHT (OA = 99.03%). (e) MASSFormer (OA = 99.27%). (f) DCTN (OA = 98.14%). (g) LSFAT (OA = 98.85%). (h) MambaHSI (OA = 97.55%). (i) 3DSS-Mamba (OA = 98.78%). (j) FS-CGNet (OA = 99.91%).

TABLE VIII

EVALUATION OF DIFFERENT METHODS ON THE XIONG AN DATASET BASED ON CLASS ACCURACY (%), OA (%), AA (%), AND κ (%)

NO.	SSFTT	morphFormer	DBCT	GAHT	MASSFormer	DCTN	LSFAT	MambaHSI	3DSS-Mamba	FS-CGNet
1	98.78	98.51	98.98	98.75	98.23	97.47	99.67	97.70	99.05	**100.0**
2	99.51	98.55	99.40	99.30	99.47	99.20	99.81	98.91	99.42	**99.99**
3	98.63	99.74	97.77	97.78	95.97	98.41	98.85	98.11	99.23	**100.0**
4	99.96	99.77	99.93	99.89	99.99	99.86	99.90	99.85	99.95	**100.0**
5	99.34	98.93	98.88	99.05	98.98	98.76	99.82	98.37	99.08	**99.99**
6	99.55	99.45	99.70	99.50	99.68	99.41	99.61	98.99	98.58	**100.0**
7	99.83	99.80	99.72	99.65	99.84	98.97	99.75	99.87	99.97	**100.0**
8	99.77	99.76	99.87	99.78	99.64	99.66	99.90	99.56	99.89	**100.0**
9	99.85	99.93	99.74	99.82	99.90	99.71	99.85	99.44	99.84	**100.0**
10	99.95	99.88	99.93	99.90	99.77	99.86	99.98	99.82	99.97	**100.0**
11	97.61	87.89	96.77	96.72	93.56	99.18	91.65	98.10	99.45	**100.0**
12	97.14	95.94	96.02	95.31	94.18	94.52	99.23	90.36	92.18	**99.89**
13	99.12	98.87	98.57	98.66	99.12	97.49	99.60	97.25	97.97	**99.98**
14	96.28	96.95	94.80	93.32	90.58	89.53	98.03	86.02	93.50	**99.95**
15	95.99	91.84	95.91	93.69	97.08	92.95	98.29	89.88	95.73	**99.91**
16	92.95	77.84	90.35	91.11	91.45	83.42	93.69	85.33	88.47	**99.88**
17	95.86	76.85	81.36	98.29	76.04	**100.0**	89.85	**100.0**	92.13	99.85
18	99.31	99.68	99.06	98.82	98.77	98.84	99.73	97.84	99.43	**99.99**
19	98.23	97.58	97.99	96.86	95.61	95.17	98.47	95.60	91.20	**100.0**
20	98.19	98.96	99.21	97.99	97.80	98.82	98.33	98.42	98.70	**99.99**
OA (%)	99.19	98.70	98.91	98.80	98.91	98.22	99.59	97.84	98.56	**99.98**
AA (%)	98.29	95.84	97.19	97.71	96.28	97.06	98.20	96.47	97.19	**99.97**
κ (%)	99.05	98.49	98.73	98.61	98.74	97.94	99.53	97.49	98.33	**99.98**

feature map. This complexity arises from operations like Fourier transforms, which involve logarithmic scaling with respect to the input size. The MSFAM has a complexity of $O(L)$, as it performs attention operations across multiple scales, with the computational cost scaling linearly with the input size. Similarly, the CGAM also has a complexity of $O(L)$, as it computes global attention over the input feature map, which scales linearly with the size of the input.

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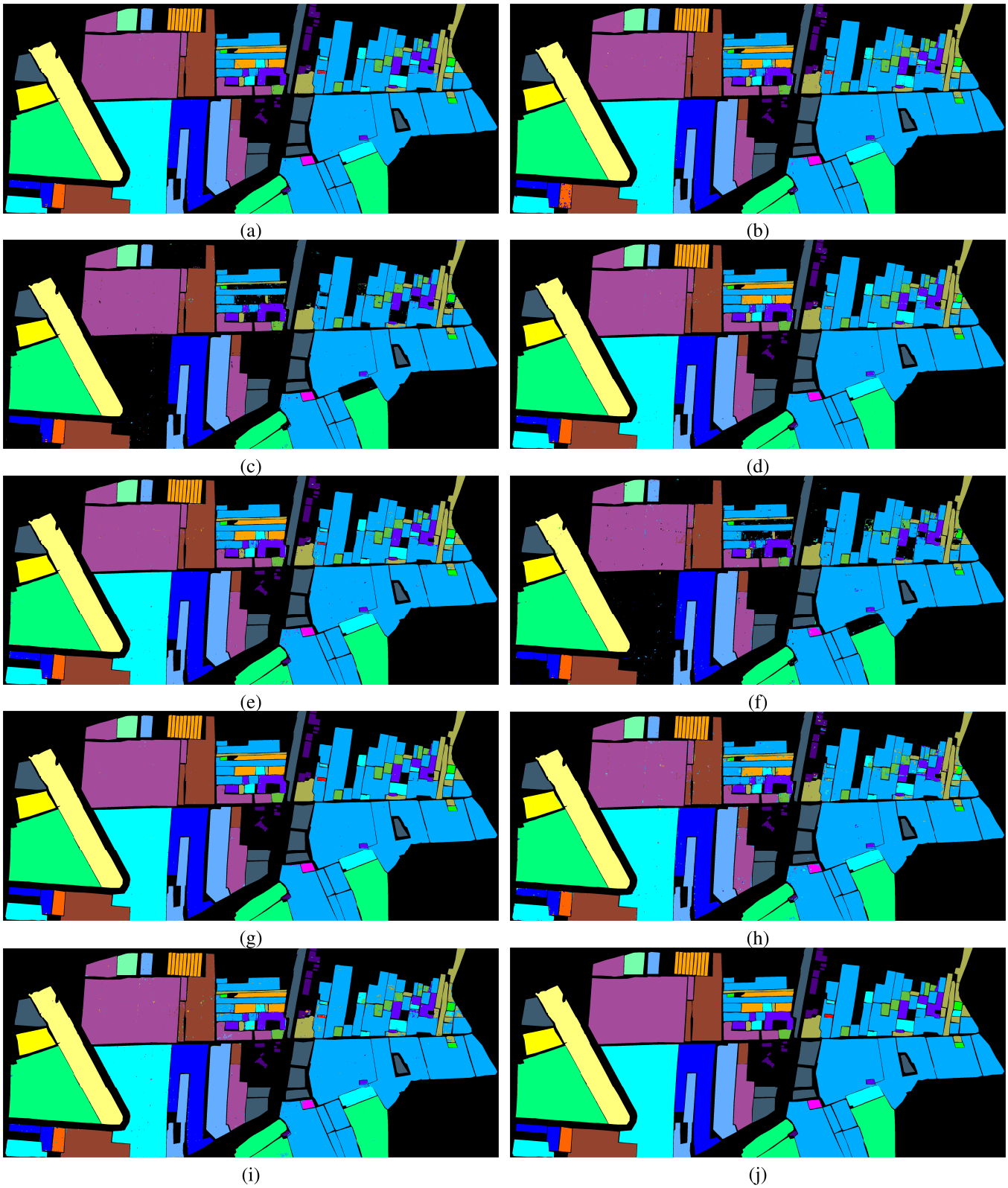


Fig. 12. Comparison of classification results on the Xiong An dataset using various methods with only 10% of the training samples. (a) SSFTT (OA = 99.19%). (b) morphFormer (OA = 98.70%). (c) DBCT (OA = 98.91%). (d) GAHT (OA = 98.80%). (e) MASSFormer (OA = 98.91%). (f) DCTN (OA = 98.22%). (g) LSFAT (OA = 99.59%). (h) MambaHSI (OA = 97.84%). (i) 3DSS-Mamba (OA = 98.56%). (j) FS-CGNet (OA = 99.98%).

D. Parameter Analysis

1) Effect of Training Set Size on OA: Fig. 14 demonstrates the variation in classification accuracy across ten different algorithms when applied to four datasets with varying proportions of training samples. The algorithms tested include SSFTT, morphFormer, DBCT, GAHT, MASSFormer, DCTN,

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Fig. 13. Confusion matrix using the proposed method on the Pavia University dataset.

LSFAT, MambaHSI, 3DSS-Mamba, and FS-CGNet, the last of which is a novel method proposed in this article. Among these, FS-CGNet performs exceptionally well across all four datasets. With training sample proportions ranging from 2% to 10%, FS-CGNet consistently maintains a high level of OA, AA, and κ, showcasing its remarkable generalization ability and stability.

First, on the Pavia University dataset, although the performance of each algorithm differs, the overall trend is that as the training sample proportion increases, the accuracy of all algorithms significantly improves. For example, FS-CGNet achieves an OA of 95.36% with 2% of the training samples, which increases to 99.83% with 10%, demonstrating FS-CGNet's excellent performance with limited labeled data. In comparison, algorithms like LSFAT and 3DSS-Mamba show relatively poor performance at lower sample proportions, but their accuracy increases significantly as the proportion of training samples increases.

Next, on the Salinas dataset, FS-CGNet also exhibits outstanding performance. Even with a lower proportion of training samples, FS-CGNet maintains high accuracy, which gradually stabilizes and improves as the sample size increases. On the other hand, algorithms such as MambaHSI and 3DSS-Mamba perform better at higher sample proportions but are less stable at lower proportions, suggesting that they rely heavily on large datasets.

On the WHU-Hi-LongKou dataset, FS-CGNet and DBCT stand out, with FS-CGNet achieving an OA of 99.91% and DBCT at 99.36% with 10% training sample proportion, highlighting their strengths in high-dimensional data processing. In contrast, DCTN and MambaHSI show lower accuracy at low sample proportions, especially at 2% and 4%, with DCTN achieving OAs of 91.18% and 93.44%, respectively, and MambaHSI achieving OAs of 91.56% and 93.83%, respectively. This suggests that DCTN and MambaHSI may struggle to effectively extract features with limited training data, which impacts accuracy.

Finally, in the Xiong An dataset, FS-CGNet continues to outperform all other algorithms at every training proportion, achieving an impressive OA of 99.98% at 10%. The LSFAT and SSFTT algorithms, while effective at high sample proportions, struggle with low sample ratios, suggesting their reliance on more data for feature extraction.

In summary, FS-CGNet consistently exhibits excellent performance across all datasets and training proportions, especially in terms of classification ability with small sample sizes. In contrast, other algorithms are more unstable at lower training sample proportions, suggesting their greater dependence on the amount of data. FS-CGNet, with its efficient feature extraction and strong learning ability, maintains high accuracy when processing complex datasets, showing excellent generalization ability and stability.

2) Effect of Patch Size on OA, AA, and κ Accuracy Metrics: The selection of patch size has a crucial impact on model performance. When the patch size is set too small, the model may not gather enough information during the training process, which can limit its learning capability and thus affect its performance. On the other hand, if the patch size is too large, it may lead to information redundancy, which increases the risk of overfitting and can potentially cause a decrease in accuracy. Furthermore, larger patch sizes not only affect the model's generalization ability but also significantly increase computational complexity, consuming more resources and time. The patch size is highly sensitive to the characteristics of the dataset and the task involved. For tasks requiring fine-grained local features (e.g., texture recognition), smaller patch sizes (e.g., 9×9 and 11×11) are generally more effective. In contrast, for tasks that require a broader global context (e.g., image classification), larger patch sizes (e.g., 17×17 and 19×19) tend to perform better. A general guideline is to experiment with varying patch sizes and evaluate the impact on classification accuracy, training stability, and generalization performance. Keep in mind that larger patch sizes also require more memory and computational resources, which should be considered based on the available hardware. Fig. 15 illustrates the impact of different patch sizes on the accuracy metrics (OA, AA, and κ) across four datasets, with patch sizes of $\{9 \times 9, 11 \times 11, 13 \times 13, 15 \times 15, 17 \times 17, 19 \times 19, 21 \times 21\}$. Each data point represents the average result from five experimental runs. From the figure, it is clear that the model achieves the best classification accuracy for the four datasets at specific patch sizes: 15×15, 17×17, 15×15, and 17×17.

3) Effect of Learning Rate on OA, AA, and κ Accuracy Metrics: The learning rate is a crucial hyperparameter that controls the step size for parameter updates during each iteration, significantly influencing both the convergence speed and stability of the model. If the learning rate is set too high, the model may experience oscillations or fail to converge during training, potentially leading to training failure. On the other hand, if the learning rate is set too low, the model's convergence becomes excessively slow, resulting in longer training times and reduced efficiency. Therefore, selecting an appropriate learning rate is critical to ensuring efficient training and stable convergence of the model. The learning

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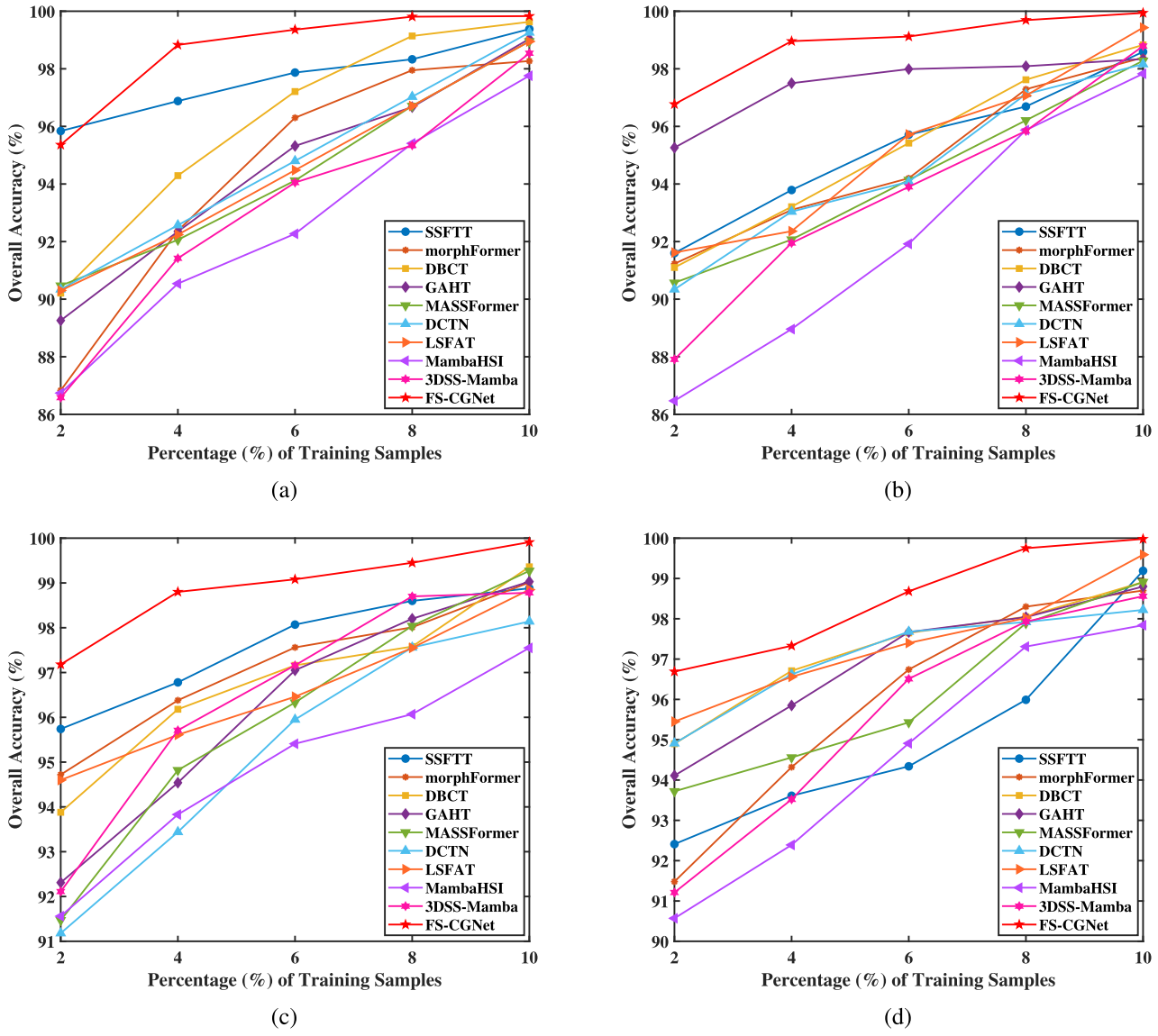


Fig. 14. Effect of different training set sizes on the performance metrics across various datasets. (a) Pavia University dataset. (b) Salinas dataset. (c) WHU-Hi-LongKou dataset. (d) Xiong An dataset.

TABLE IX

TRAINING TIME IN MINUTES, TEST TIME IN SECONDS, AND FLOPS FOR THE CONTRAST METHODS AND THE PROPOSED METHOD ACROSS FOUR DATASETS

Methods	Pavia University			Salinas			WHU-Hi-LongKou			Xiong An		
	Train (m)	Test (s)	FLOPs	Train (m)	Test (s)	FLOPs	Train (m)	Test (s)	FLOPs	Train (m)	Test (s)	FLOPs
SSFTT	0.92	1.02	63.50M	1.18	1.86	73.51M	10.50	20.98	90.21M	69.65	127.20	139.03M
morphFormer	0.94	1.13	70.02M	1.20	2.07	84.10M	10.93	21.91	97.18M	75.17	157.24	148.24M
DBCT	0.89	1.10	66.95M	1.17	1.47	77.12M	10.81	21.18	95.18M	73.78	128.90	147.11M
GAHT	1.67	1.46	77.55M	2.31	2.45	98.96M	13.46	25.16	137.52M	78.80	148.89	164.14M
MASSFormer	1.15	1.26	72.50M	1.30	2.38	94.56M	12.76	24.13	128.17M	82.86	146.12	154.09M
DCTN	5.17	2.40	207.20M	8.21	3.92	227.15M	21.21	37.50	264.57M	97.18	182.90	397.18M
LSFAT	1.23	1.25	76.85M	1.26	2.08	95.32M	12.21	23.51	132.37M	77.05	144.56	156.13M
MambaHSI	4.57	2.13	151.18M	5.19	2.31	174.56M	15.43	28.75	237.12M	88.31	163.43	289.11M
3DSS-Mamba	3.66	1.77	139.03M	4.09	3.01	163.45M	13.63	25.40	202.14M	80.23	152.32	267.89M
FS-CGNet	1.04	1.17	72.40M	1.23	2.00	94.19M	11.39	21.81	126.63M	74.46	139.56	150.84M

rate is highly sensitive to both the dataset and the classification architecture. For complex datasets or deep architectures, a smaller learning rate (e.g., 0.0001–0.0003) can help avoid overshooting the optimal solution and ensure smooth convergence. For simpler datasets or shallower models, a higher learning rate (e.g., 0.0004–0.0005) may accelerate

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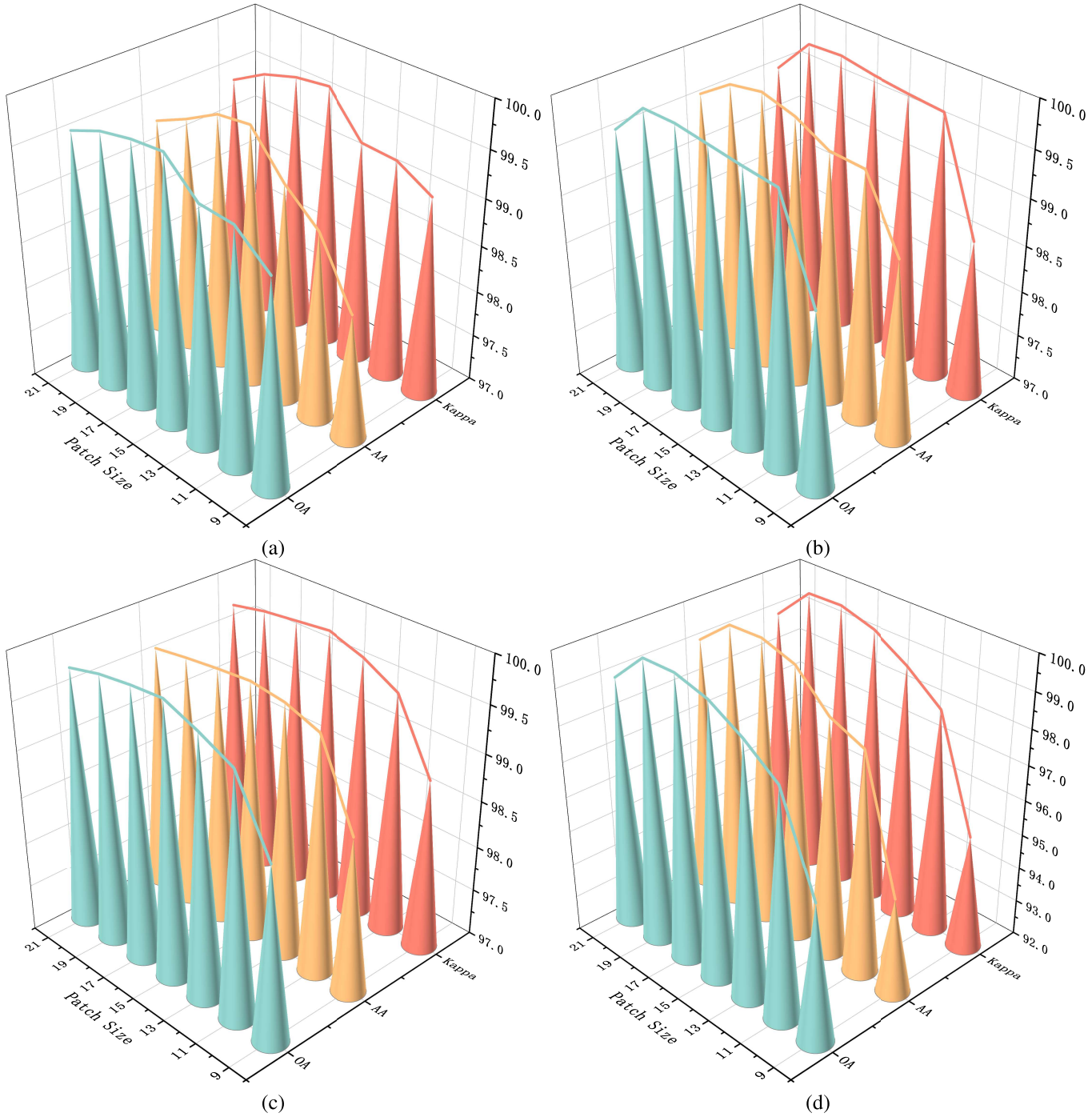


Fig. 15. Effect of different patch sizes on the performance metrics across various datasets. (a) Pavia University dataset. (b) Salinas dataset. (c) WHU-Hi-LongKou dataset. (d) Xiong An dataset.

convergence without sacrificing performance. Fig. 16 illustrates the variation in accuracy across four datasets within the learning rate range of $\{0.0001, 0.0002, 0.0003, 0.0004, 0.0005\}$. From the figure, it is evident that the optimal accuracy for the four datasets is achieved at learning rates of 0.0002, 0.0004, 0.0003, and 0.0003. These learning rates provide the best training performance for each corresponding dataset.

E. Computational Complexity Analysis

To systematically evaluate the computational complexity of various methods, the experimental results in Table IX are provided from three quantitative indices: training time, testing time, and floating-point operations (FLOPs).

Among the Transformer-based methods, lightweight models such as SSFTT, DBCT, and FS-CGNet demonstrate relatively low computational overhead across four datasets. Their test times are generally short, especially for the Pavia University dataset, and FLOPs range from 63.50 to 150.84 M, reflecting good computational efficiency and broad applicability. In contrast, DCTN's training time, testing time, and FLOPs are markedly higher than those of other methods across the four datasets, reflecting its structural complexity

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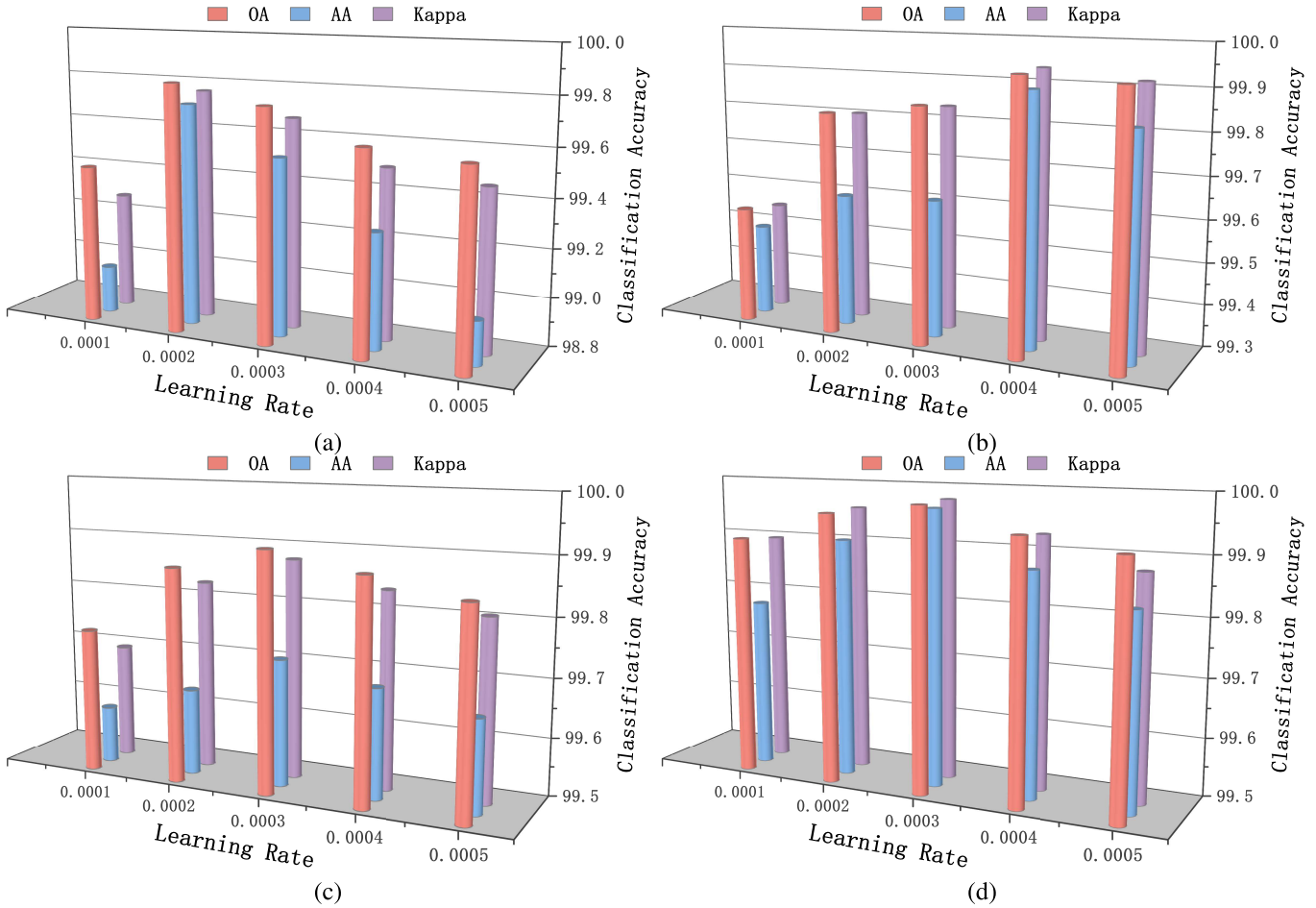


Fig. 16. Effect of different learning rates on the performance metrics across various datasets. (a) Pavia University dataset. (b) Salinas dataset. (c) WHU-Hi-LongKou dataset. (d) Xiong An dataset.

and higher computational resource demands. As a dual-branch network combining local convolution and global Transformer attention, DCTN fully exploits spatial details and global spectral information but also results in substantial computational cost.

The Mamba series methods (MambaHSI and 3DSS-Mamba) employ a linear-complexity state-space model to mitigate the quadratic computational bottleneck of Transformer-based approaches. However, as shown in Table IX, these methods still incur considerable training time, testing time, and FLOPs across all four datasets. This is primarily due to MambaHSI's use of multiple feature extraction modules (spatial Mamba block, spectral Mamba block, and spatial–spectral fusion) for comprehensive spatial and spectral feature extraction. Similarly, 3DSS-Mamba integrates several strategies, including spectral–spatial token generation and a 3-D-spectral–spatial selective scanning mechanism, to capture global semantic information from high-dimensional data.

In comparison, the proposed FS-CGNet achieves a brilliant balance between accuracy and efficiency across all datasets. It can be seen from Table IX that its training and testing times consistently remain at low levels, and its FLOPs are lower than those of the vast majority of comparison methods. The experimental results in Tables V–VIII demonstrate that FS-CGNet leads in classification accuracy compared with other comparative algorithms, and similar results are also presented in Fig. 14. Overall, FS-CGNet delivers strong competitive performance in terms of classification accuracy, training time, testing time, and FLOPs.

F. Ablation Experiment

In order to comprehensively evaluate the impact of each module in the proposed model, a series of ablation experiments is conducted on four different datasets. The model consists of three key components: FDSFM, MSFAM, and CGAM. The results from these experiments, presented in Table X, provide a detailed analysis and evaluation of the model's performance based on three classification metrics: OA, AA, and κ.

First, in case 1, only the FDSFM is used. The FDSFM enhances the model's understanding of hyperspectral data by extracting local and global structural features from the frequency domain, thereby improving the classification performance. However, due to the lack of support from other modules, the performance improvement achieved by FDSFM is relatively limited. The OA of the Pavia University dataset is 77.86%, the AA is 73.67%, and κ is 75.79%. The OA

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TABLE X
ABLATION EXPERIMENT RESULTS ON PAVIA UNIVERSITY, SALINAS, WHU-HI-LONGKOU, AND XIONG AN DATASETS

Cases	Components			Indicators											
				Pavia University			Salinas			WHU-Hi-LongKou			Xiong An		
	FDSFM	MSFAM	CGAM	OA(%)	AA(%)	κ(%)	OA(%)	AA(%)	κ(%)	OA(%)	AA(%)	κ(%)	OA(%)	AA(%)	κ(%)
1	✓	×	×	77.86	73.67	75.79	76.58	72.28	75.05	77.14	73.41	76.23	76.30	74.31	75.71
2	×	✓	×	75.62	74.12	74.19	76.71	75.53	76.34	79.69	79.29	79.60	78.79	76.76	78.59
3	×	×	✓	86.20	83.67	85.88	88.89	85.45	87.06	89.29	88.57	89.06	89.65	88.84	89.60
4	✓	✓	×	92.76	91.57	92.35	92.39	90.33	91.51	92.72	90.09	92.32	92.50	91.80	92.42
5	✓	×	✓	91.65	90.35	90.88	93.17	92.05	93.08	92.95	90.73	90.65	93.71	92.68	93.50
6	×	✓	✓	95.02	94.06	95.55	95.51	94.46	95.01	96.99	95.39	96.53	97.68	96.18	97.47
7	✓	✓	✓	**99.83**	**99.73**	**99.77**	**99.94**	**99.90**	**99.94**	**99.91**	**99.72**	**99.88**	**99.98**	**99.97**	**99.98**

of the Salinas dataset is 76.58%, the AA is 72.28%, and κ is 75.05%. These results indicate that although FDSFM can obtain a certain level of accuracy, a single module fails to achieve the best performance when processing complex HSIs. In case 2, only the MSFAM is used. The MSFAM enhances the feature expression of spectral channels at different scales through the attention mechanism, highlighting important spectral information. The experimental results show that the OA of the Salinas dataset has reached 76.71%, AA has reached 75.53%, and κ has reached 76.34%. This indicates that the fine processing of feature selection by MSFAM helps the model better focus on key spectral channels. However, models that still rely solely on a single module fail to sufficiently exploit the global information of images. In case 3, the CGAM is used, mainly enhancing the expression of global features through the aggregation of multiscale information. The experimental results show that the model using the CGAM performs better than cases 1 and 2 on multiple datasets. Especially on the WHU-Hi-LongKou dataset, OA reaches 89.29%, AA reaches 88.57%, and κ reaches 89.06%.

Second, in case 4, the FDSFM and MSFAM work together. The experimental results show that the model performance has been further improved compared with cases 1 and 2. Especially on the Pavia University dataset, OA has reached 92.76%, AA has reached 91.57%, and κ has reached 92.35%. This indicates that the combination of FDSFM and MSFAM is effective in improving classification accuracy. Next, in case 5, the FDSFM and CGAM are used in combination. The results show that the classification accuracy of the model is good on multiple datasets, especially on the Xiong An dataset, where OA has increased to 93.71%, AA has increased to 92.68%, and κ has increased to 93.50%. This result indicates that the FDSFM, on the basis of extracting frequency-domain features, combined with the global information aggregation of the CGAM, enhances the model's processing ability for discriminating different land cover classes. In case 6, the MSFAM and CGAM are used in combination. The experimental results still demonstrated significant performance improvements, especially on the WHU-Hi-LongKou dataset, where OA is 96.99%, AA is 95.39%, and κ is 96.53%. This indicates that the combination of the two modules plays a key role in improving classification accuracy.

Finally, in case 7, all three modules are used. The experimental results show that the model achieves the best performance on all datasets. These results indicate that by jointly using all modules, the model can significantly enhance its overall classification ability through the extraction and fusion of multiscale and multifrequency features.

IV. DISCUSSION

Tables V–VIII show that FS-CGNet performs best in terms of OA, AA, and Kappa coefficient. The comparison algorithms include two categories: Transformer-based SSFTT, morphFormer, DBCT, GAHT, MASSFormer, DCTN, and LSFAT, as well as the Mamba-based MambaHSI and 3DSS-Mamba. In scenarios with limited training samples, Transformer-based methods are prone to overfit due to the existing large number of parameters. While the Mamba-based method focuses on global receptive fields, it exhibits some limitations in handling fine-grained spectral–spatial joint representations, which results in reduced discriminative ability under small sample conditions. To address these challenges, FS-CGNet effectively leverages its novel modules. The FDSFM utilizes frequency-domain information to capture multiscale and multifrequency features, significantly enhancing the model's capability to handle high-dimensional data and alleviating overfitting issues in small sample settings. Subsequently, the MSFAM adaptively enhances the weights of critical spectral channels, further improving classification accuracy and robustness. Finally, the CGAM aggregates features across different scales, enabling global optimization and substantially boosting classification performance.

The implementation of FS-CGNet faces challenges due to data noise, environmental variations, limited computational resources, and time constraints. HSIs are vulnerable to atmospheric disturbances, sensor noise, and lighting changes, which can degrade image quality and affect model performance. Future work will focus on improving data preprocessing with noise removal, radiometric and atmospheric correction, as well as data augmentation techniques like random occlusion and spectral perturbation to enhance model robustness in complex environments.

Additionally, hyperspectral data's high volume and dimensionality increase computational burdens, prolonging training and inference times, which challenges real-time applications like urban planning and disaster monitoring. Future work will explore GPU parallel acceleration, model compression techniques such as parameter quantization, network pruning, and knowledge distillation to enhance computational efficiency,

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reduce resource consumption, and ensure rapid deployment of FS-CGNet without compromising accuracy.

V. Conclusion

In this article, we propose the FS-CGNet framework, which effectively enhances the HSI classification performance by integrating frequency-domain spatial–spectral feature extraction, attention-driven multiscale spectral–channel fusion, and cross-scale global aggregation. The experimental results demonstrate that FS-CGNet leverages multiscale and multifrequency information in hyperspectral data, significantly improving classification accuracy, particularly in complex tasks, and outperforming existing state-of-the-art methods. The FDSFM module captures both local and global structural features by extracting frequency-domain information. The MSFAM module dynamically emphasizes the importance of critical spectral channels using an attention mechanism, improving classification robustness; the CGAM module aggregates information across multiple scales, optimizing the global representation of the model and minimizing the misclassification of pixels as much as possible. Experiments on four benchmark hyperspectral datasets show that FS-CGNet obtains better classification performance comparable to some leading methods in the field, demonstrating its potential in HSI classification tasks. Future work may further explore the scalability and optimization of FS-CGNet, particularly its capability to handle large-scale hyperspectral datasets in practical applications.

The proposed FS-CGNet model effectively enhances the ability to capture both local and global patterns in HSI classification through the integration of the FDSFM, MSFAM, and CGAM modules. However, the model still faces certain limitations. Although the FDSFM, MSFAM, and CGAM modules significantly enhance classification accuracy, the computational consumption also increases to a certain extent, especially when processing large-scale hyperspectral data. To address this challenge and reduce computational complexity, future work aims to introduce intelligent optimization algorithms and lightweight models, such as pruning and quantization methods applied to CNNs and attention mechanisms, which are expected to enhance the training speed and efficiency of the model while maintaining or potentially improving its classification accuracy.

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